DATABASE MANAGEMENT SYSTEMS · PROJECT 59

Design and Implementation of a Database Management System for Police Complaint and Evidence Tracking

PCETS — Police Complaint & Evidence Tracking System

Database	MySQL 8.0 (InnoDB) — schema in 3NF
Application	Python 3 · Flask web application
Deliverables	Review 1, Review 2 and Review 3
Report generated	16 October 2026

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1. Introduction

1.1 Problem identification

Police stations record complaints (FIRs), investigation diaries, seized evidence, court dates and custody hand-overs in separate paper registers. Such unstructured registers make it hard to answer basic questions — *Which cases are pending and with whom? Where is exhibit X right now? Who held it on a given date?* — and, most importantly, they weaken the **chain of custody**: a missing or overlapping hand-over entry can make evidence inadmissible in court.

1.2 Scope

PCETS is a relational database and web application for one police district (several stations). It covers complaint registration, officer assignment, the investigation diary, evidence registration and custody transfers, witnesses, accused persons, charges, court references, case status workflow with a full audit trail, search and reporting. Court case management beyond references/hearing dates, payroll and FIR filing with external systems (CCTNS) are out of scope.

1.3 Objectives

- Design a normalized (3NF) relational schema for complaints, persons, officers, assignments, activities, evidence, custody logs, witnesses, charges, court references and status histories.
- Enforce the business rules **inside the database** — unique complaint/evidence IDs, complete custody transfer, valid case status, no overlapping custody and mandatory closure details — using keys, CHECK constraints and triggers.
- Provide views and reports: pending cases, officer load, evidence location, custody history, witnesses, case timeline and closures.
- Build a secure, validated, transaction-aware application with role-based access for station staff.
- Demonstrate correctness through systematic automated testing.

2. Users and roles

Role	Typical user	Main permissions
ADMIN	District system administrator	Everything, including officer management and the Integrity Lab
SHO	Station House Officer (Inspector)	Register & edit complaints, assign/relieve officers, change status, **close** cases, all case actions, reports
IO	Investigating Officer (Sub-Inspector)	Register complaints, log activities, evidence, custody transfers, witnesses, accused, charges, court references, status updates (not closure)
CUSTODIAN	Malkhana in-charge (Head Constable)	Register evidence and transfer custody; read-only access to cases and reports

Civilian **persons** (complainants, accused, witnesses) are data subjects, not system users. There is no login page: the application opens as the default user (`default_user` in `config.ini`, normally `admin`) and the **Acting as** selector in the sidebar switches to any station user. Permissions for the acting role are enforced on the server for every request (HTTP 403 otherwise), the UI hides actions the role cannot perform, and the acting officer is recorded in the audit trail.

3. Requirements

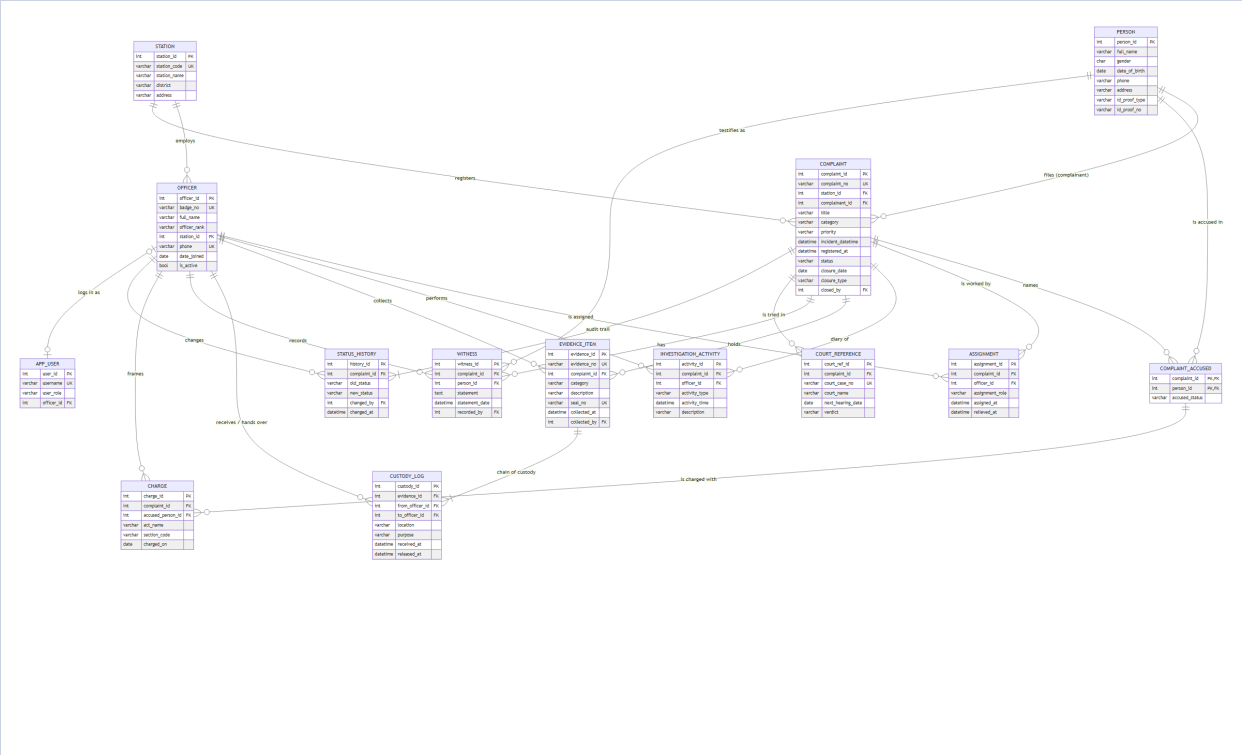
3.1 Functional requirements

ID	Requirement	Implemented in
FR-01	Register a complaint with an existing or new complainant; issue a unique complaint number	Register complaint page; trigger trg_complaint_bi
FR-02	Assign / relieve officers with roles (LEAD, SUPPORT, FORENSIC, SURVEILLANCE); one active lead per case	Case → Officers tab
FR-03	Log investigation activities (site visit, raid, arrest, statement …) by assigned officers	Case → Investigation diary
FR-04	Register evidence with seal number; start chain of custody automatically	Evidence → Register
FR-05	Transfer custody between officers with purpose, location, time and seal check	Evidence → Transfer custody
FR-06	Record witnesses (with protected-witness flag) and accused persons	Case → Accused & witnesses
FR-07	Frame charges (act + section) against accused; add court references and hearings	Case → Charges & court
FR-08	Update case status through a valid workflow; closure requires details; re-open allowed	Update status dialog; trg_complaint_bu
FR-09	Maintain a tamper-proof status history and full case timeline	status_history, v_case_timeline
FR-10	Search complaints, evidence, persons and officers; filter lists	Global search, list filters
FR-11	Reports: pending cases, officer load, evidence location, custody history, witnesses, case timeline, closures (+ analytics)	Reports module, CSV export, print
FR-12	CRUD for persons and officers with referential protection	Persons, Officers modules
FR-13	Role-based access with an acting-user selector (no login page)	auth.py, PERMISSIONS map

3.2 Non-functional requirements

- **Integrity:** every business rule is enforced by MySQL so it holds even if the application is bypassed.
- **Atomicity:** multi-step operations (registration, custody transfer, closure, lead replacement) run as single transactions.
- **Security:** role-based access for the acting user, CSRF tokens, parameterised SQL only.
- **Auditability:** custody log is append-only, status history cannot be edited or deleted, complaints cannot be deleted.
- **Performance:** indexes on all frequently searched/joined columns; views pre-join common report data.
- **Usability:** responsive UI (desktop and mobile), light/dark theme, field-level error messages, printable case file and custody sheet.

4. Conceptual design (ER model)



Crow's-foot ER diagram generated from `docs/er_diagram.mmd`. PK = primary key, FK = foreign key, UK = unique key.

4.1 Entities

Entity	Description
STATION	Police station / unit that registers complaints and employs officers
OFFICER	Police personnel with badge, rank and station; may have an app user account
PERSON	Any civilian — the same person can be complainant, accused or witness in different cases
COMPLAINT	The case (FIR) — strong entity at the centre of the model, with status and closure details
COMPLAINT_ACCUSED	Associative entity: persons accused in a complaint, with arrest/bail status
ASSIGNMENT	Associative entity with history: which officer works on which case, role and period
INVESTIGATION_ACTIVITY	Diary entry of an action taken in a case
EVIDENCE_ITEM	Seized / collected exhibit belonging to a complaint
CUSTODY_LOG	One custody period of an evidence item: who held it, where, why, from when to when
WITNESS	Statement of a person in a complaint
CHARGE	Legal section framed against an accused of a complaint
COURT_REFERENCE	Court case linked to a complaint with hearing date and verdict
STATUS_HISTORY	Audit trail of every status change of a complaint
APP_USER	Application user and role (selected via Acting as)

4.2 Relationships and cardinalities

Relationship	Cardinality	Notes
Station employs Officer	1 : N	every officer belongs to exactly one station
Station registers Complaint	1 : N	
Person files Complaint	1 : N	complainant
Complaint — Person (accused)	M : N	resolved by COMPLAINT_ACCUSED
Complaint — Officer (assignment)	M : N	resolved by ASSIGNMENT, keeps history
Complaint has Activities	1 : N	performed by one officer each
Complaint holds Evidence	1 : N	collected by one officer
Evidence has Custody periods	1 : N (≥1)	total participation — every item has an initial seizure

Complaint — Person (witness)	M : N	resolved by WITNESS
Complaint_Accused is charged with Charge	1 : N	composite FK (complaint_id, person_id)
Complaint is tried in Court reference	1 : N	
Complaint has Status history	1 : N (≥1)	first row written on registration
Officer logs in as App user	1 : 0..1	ADMIN may have no officer

5. Relational schema

Primary keys are underlined, foreign keys are shown in blue as *column → table*.

STATION(**station_id**, station_code UK, station_name, district, address, phone)

OFFICER(**officer_id**, badge_no UK, full_name, officer_rank, station_id → STATION, phone UK, email UK, date_joined, is_active)

PERSON(**person_id**, full_name, gender, date_of_birth, phone, email, address, id_proof_type, id_proof_no, created_at) UK(id_proof_type, id_proof_no)

COMPLAINT(**complaint_id**, complaint_no UK, station_id → STATION, complainant_id → PERSON, title, description, category, priority, incident_datetime, incident_location, registered_at, registered_by → OFFICER, status, closure_date, closure_type, closure_remarks, closed_by → OFFICER)

COMPLAINT_ACCUSED(**complaint_id → COMPLAINT, person_id → PERSON**, accused_status, added_at, remarks)

ASSIGNMENT(**assignment_id**, complaint_id → COMPLAINT, officer_id → OFFICER, assignment_role, assigned_at, relieved_at, assigned_by → OFFICER, remarks)

INVESTIGATION_ACTIVITY(**activity_id**, complaint_id → COMPLAINT, officer_id → OFFICER, activity_type, activity_time, location, description, outcome, logged_at)

EVIDENCE_ITEM(**evidence_id**, evidence_no UK, complaint_id → COMPLAINT, category, description, quantity, item_condition, seal_no UK, collected_at, collected_by → OFFICER, collected_from, registered_at)

CUSTODY_LOG(**custody_id**, evidence_id → EVIDENCE_ITEM, from_officer_id → OFFICER, to_officer_id → OFFICER, location, purpose, received_at, released_at, seal_intact, remarks)

WITNESS(**witness_id**, complaint_id → COMPLAINT, person_id → PERSON, statement, statement_date, recorded_by → OFFICER, is_protected) UK(complaint_id, person_id)

CHARGE(**charge_id**, (complaint_id, accused_person_id) → COMPLAINT_ACCUSED, act_name, section_code, offence_description, charged_on, charged_by → OFFICER)

COURT_REFERENCE(**court_ref_id**, complaint_id → COMPLAINT, court_case_no UK, court_name, court_type, filing_date, next_hearing_date, judge_name, case_stage, verdict)

STATUS_HISTORY(**history_id**, complaint_id → COMPLAINT, old_status, new_status, changed_by → OFFICER, changed_at, remarks)

APP_USER(**user_id**, username UK, password_hash, user_role, officer_id → OFFICER UK, is_active, last_login, created_at)

> Surrogate integer keys are used everywhere for stable joins; the human-readable identifiers (`complaint_no`, `evidence_no`, `badge_no`, `court_case_no`) are candidate keys protected by UNIQUE constraints.

6. Functional dependencies and normalization

6.1 Functional dependencies

Relation	Functional dependencies (non-trivial)
STATION	station_id → station_code, station_name, district, address, phone; station_code → station_id
OFFICER	officer_id → badge_no, full_name, officer_rank, station_id, phone, email, date_joined, is_active; badge_no → officer_id; phone → officer_id
PERSON	person_id → all attributes; (id_proof_type, id_proof_no) → person_id
COMPLAINT	complaint_id → complaint_no, station_id, complainant_id, title, description, category, priority, incident_datetime, incident_location, registered_at, registered_by, status, closure_*; complaint_no → complaint_id
COMPLAINT_ACCUSED	(complaint_id, person_id) → accused_status, added_at, remarks
ASSIGNMENT	assignment_id → complaint_id, officer_id, assignment_role, assigned_at, relieved_at, assigned_by, remarks
INVESTIGATION_ACTIVITY	activity_id → complaint_id, officer_id, activity_type, activity_time, location, description, outcome, logged_at
EVIDENCE_ITEM	evidence_id → evidence_no, complaint_id, category, description, quantity, item_condition, seal_no, collected_*; evidence_no → evidence_id; seal_no → evidence_id
CUSTODY_LOG	custody_id → evidence_id, from_officer_id, to_officer_id, location, purpose, received_at, released_at, seal_intact, remarks; (evidence_id, received_at) → custody_id (guaranteed by the no-overlap rule)
WITNESS	witness_id → complaint_id, person_id, statement, statement_date, recorded_by, is_protected; (complaint_id, person_id) → witness_id
CHARGE	charge_id → complaint_id, accused_person_id, act_name, section_code, offence_description, charged_on, charged_by; (complaint_id, accused_person_id, act_name, section_code) → charge_id
COURT_REFERENCE	court_ref_id → all; court_case_no → court_ref_id
STATUS_HISTORY	history_id → complaint_id, old_status, new_status, changed_by, changed_at, remarks
APP_USER	user_id → all; username → user_id; officer_id → user_id

6.2 Normal forms

- **1NF** — every attribute is atomic: there are no repeating groups (multiple accused, witnesses, charges, custody steps and status changes are stored as rows in their own tables, never as lists in a column).
- **2NF** — the only composite primary key is COMPLAINT_ACCUSED(complaint_id, person_id); its non-key attributes (accused_status, added_at, remarks) describe the *pair*, not either part alone, so there is no partial dependency. All other tables have single-column keys.
- **3NF** — in every relation each determinant of a non-trivial FD is a (candidate) key, so there are no transitive dependencies (the schema is in fact in **BCNF**).

6.3 Normalization decisions

Un-normalized design (register style)	Problem	Decision
evidence_item(current_holder, current_location, evidence_status)	Transitive dependency on the latest custody row; update anomaly — moving an item means updating two places	Holder, location and status are **derived** from the open CUSTODY_LOG row in view `v_evidence_location`
complaint(officer_name, officer_rank, officer_station)	officer_id → name, rank (transitive), and only one officer per case	ASSIGNMENT table (M:N with history); officer details stored once in OFFICER
complaint(complainant_name, phone, address …)	Same person repeated in many cases, inconsistent updates	Single PERSON table referenced by complaint, accused and witness
complaint(accused1, accused2, sections …)	Repeating groups (violates 1NF)	COMPLAINT_ACCUSED and CHARGE tables
charge(complaint_id, person_id) with separate FKs	Allowed charging someone who is not an accused of that case	Composite FK to COMPLAINT_ACCUSED — the relationship itself is referenced
complaint(status) only	History of status changes lost	STATUS_HISTORY written by trigger

7. Data dictionary

Generated from `information_schema` of database `police_evidence_db`. Key: PRI = primary key, UNI = unique, MUL = indexed/foreign key; NN = NOT NULL.

1. app_user — 11 sample rows

Column	Type	Null	Key	Default	Extra
user_id	int	NN	PK		auto_increment
username	varchar(30)	NN	UNI		
password_hash	varchar(255)	NN			
user_role	varchar(10)	NN			
officer_id	int	YES	UNI		
is_active	tinyint(1)	NN		1	
last_login	datetime	YES			
created_at	datetime	NN		CURRENT_TIMESTAMP	DEFAULT_GENERATED

2. assignment — 27 sample rows

Column	Type	Null	Key	Default	Extra
assignment_id	int	NN	PK		auto_increment
complaint_id	int	NN	MUL		
officer_id	int	NN	MUL		
assignment_role	varchar(15)	NN		SUPPORT	
assigned_at	datetime	NN		CURRENT_TIMESTAMP	DEFAULT_GENERATED
relieved_at	datetime	YES			
assigned_by	int	YES	MUL		
remarks	varchar(300)	YES			
active_key	tinyint	YES			STORED GENERATED
lead_key	tinyint	YES			STORED GENERATED

3. charge — 10 sample rows

Column	Type	Null	Key	Default	Extra
charge_id	int	NN	PK		auto_increment
complaint_id	int	NN	MUL		
accused_person_id	int	NN			
act_name	varchar(15)	NN			
section_code	varchar(20)	NN			
offence_description	varchar(255)	NN			
charged_on	date	NN			
charged_by	int	NN	MUL		

4. complaint — 20 sample rows

Column	Type	Null	Key	Default	Extra
complaint_id	int	NN	PK		auto_increment
complaint_no	varchar(20)	NN	UNI		
station_id	int	NN	MUL		
complainant_id	int	NN	MUL		
title	varchar(150)	NN	MUL		
description	text	NN			
category	varchar(30)	NN	MUL		
priority	varchar(10)	NN	MUL	MEDIUM	
incident_datetime	datetime	NN	MUL		
incident_location	varchar(255)	NN			
registered_at	datetime	NN		CURRENT_TIMESTAMP	DEFAULT_GENERATED

registered_by	int	YES	MUL		
status	varchar(25)	NN	MUL	REGISTERED	
closure_date	date	YES			
closure_type	varchar(30)	YES			
closure_remarks	varchar(500)	YES			
closed_by	int	YES	MUL		

5. complaint_accused — 10 sample rows

Column	Type	Null	Key	Default	Extra
complaint_id	int	NN	PK		
person_id	int	NN	PK		
accused_status	varchar(15)	NN		SUSPECT	
added_at	datetime	NN		CURRENT_TIMESTAMP	DEFAULT_GENERATED
remarks	varchar(300)	YES			

6. court_reference — 3 sample rows

Column	Type	Null	Key	Default	Extra
court_ref_id	int	NN	PK		auto_increment
complaint_id	int	NN	MUL		
court_case_no	varchar(30)	NN	UNI		
court_name	varchar(150)	NN			
court_type	varchar(15)	NN			
filing_date	date	NN			
next_hearing_date	date	YES	MUL		
judge_name	varchar(100)	YES			
case_stage	varchar(20)	NN		FILED	
verdict	varchar(12)	NN		PENDING	

7. custody_log — 56 sample rows

Column	Type	Null	Key	Default	Extra
custody_id	int	NN	PK		auto_increment
evidence_id	int	NN	MUL		
from_officer_id	int	YES	MUL		
to_officer_id	int	NN	MUL		
location	varchar(150)	NN			
purpose	varchar(20)	NN			
received_at	datetime	NN			
released_at	datetime	YES			
seal_intact	tinyint(1)	NN		1	
remarks	varchar(500)	YES			
open_key	tinyint	YES			STORED GENERATED

8. evidence_item — 22 sample rows

Column	Type	Null	Key	Default	Extra
evidence_id	int	NN	PK		auto_increment
evidence_no	varchar(20)	NN	UNI		
complaint_id	int	NN	MUL		
category	varchar(20)	NN	MUL		
description	varchar(500)	NN			
quantity	int	NN		1	
item_condition	varchar(12)	NN		INTACT	
seal_no	varchar(20)	YES	UNI		
collected_at	datetime	NN			

collected_by	int	NN	MUL		
collected_from	varchar(255)	NN			
registered_at	datetime	NN		CURRENT_TIMESTAMP	DEFAULT_GENERATED

9. investigation_activity — 56 sample rows

Column	Type	Null	Key	Default	Extra
activity_id	int	NN	PK		auto_increment
complaint_id	int	NN	MUL		
officer_id	int	NN	MUL		
activity_type	varchar(25)	NN			
activity_time	datetime	NN			
location	varchar(255)	YES			
description	varchar(1000)	NN			
outcome	varchar(500)	YES			
logged_at	datetime	NN		CURRENT_TIMESTAMP	DEFAULT_GENERATED

10. officer — 16 sample rows

Column	Type	Null	Key	Default	Extra
officer_id	int	NN	PK		auto_increment
badge_no	varchar(12)	NN	UNI		
full_name	varchar(100)	NN	MUL		
officer_rank	varchar(30)	NN			
station_id	int	NN	MUL		
phone	varchar(10)	NN	UNI		
email	varchar(100)	YES	UNI		
date_joined	date	NN			
is_active	tinyint(1)	NN		1	

11. person — 40 sample rows

Column	Type	Null	Key	Default	Extra
person_id	int	NN	PK		auto_increment
full_name	varchar(100)	NN	MUL		
gender	char(1)	NN			
date_of_birth	date	YES			
phone	varchar(10)	YES	MUL		
email	varchar(100)	YES			
address	varchar(255)	NN			
id_proof_type	varchar(20)	YES	MUL		
id_proof_no	varchar(20)	YES			
created_at	datetime	NN		CURRENT_TIMESTAMP	DEFAULT_GENERATED

12. station — 4 sample rows

Column	Type	Null	Key	Default	Extra
station_id	int	NN	PK		auto_increment
station_code	varchar(10)	NN	UNI		
station_name	varchar(100)	NN			
district	varchar(60)	NN			
address	varchar(255)	NN			
phone	varchar(15)	YES			

13. status_history — 52 sample rows

Column	Type	Null	Key	Default	Extra
history_id	int	NN	PK		auto_increment
complaint_id	int	NN	MUL		

old_status	varchar(25)	YES			
new_status	varchar(25)	NN			
changed_by	int	YES	MUL		
changed_at	datetime	NN		CURRENT_TIMESTAMP	DEFAULT_GENERATED
remarks	varchar(500)	YES			

14. witness — 11 sample rows

Column	Type	Null	Key	Default	Extra
witness_id	int	NN	PK		auto_increment
complaint_id	int	NN	MUL		
person_id	int	NN	MUL		
statement	text	NN			
statement_date	datetime	NN			
recorded_by	int	NN	MUL		
is_protected	tinyint(1)	NN		0	

8. Integrity constraints and business rules

8.1 Mandatory business rules

Rule	How it is enforced
R1 Unique complaint / evidence IDs	UNIQUE `uq_complaint_no`, `uq_evidence_no`; numbers generated by triggers `trg_complaint_bi` / `trg_evidence_bi` in format CMP-YYYY-NNNNN / EVD-YYYY-NNNNN; CHECK with REGEXP on the format; numbers cannot be changed (trg_complaint_bu)
R2 Complete custody transfer	NOT NULL on to_officer, location, purpose, received_at; CHECK `chk_custody_from` (every non-seizure entry names the handing officer); trigger `trg_custody_bi`: handing officer must be the current holder, previous period must be released, new period must start exactly when the previous ended (no gap); remarks mandatory in the app for broken seal / return / disposal
R3 Valid case status	CHECK `chk_complaint_status` (5 values); trigger `trg_complaint_bu` enforces the workflow (REGISTERED→UNDER_INVESTIGATION→CHARGE_SHEETED→IN_COURT→CLOSED, closure from any open state, re-open from CLOSED) and preconditions (officer assigned, ≥1 charge, court reference)
R4 No overlapping custody	UNIQUE (evidence_id, open_key) on a generated column — at most one open custody period per item; interval-overlap check in `trg_custody_bi`; append-only log (`trg_custody_bu`, `trg_custody_bd`); final RETURN/DISPOSAL ends the chain
R5 Closure details required	CHECK `chk_complaint_closure`: status CLOSED ⇔ closure_date, closure_type, closure_remarks and closed_by are all present; CHECK `chk_complaint_closuredate`: closure not before registration

8.2 Additional rules

- Only one active LEAD officer per case and no duplicate active assignment (UNIQUE on generated columns `lead_key`, `active_key`).
- Inactive officers cannot be assigned or receive custody; activities only by officers assigned at that time; nothing can be added to a closed case.
- An accused cannot be a witness (or complainant) in the same case; charges only against accused of that case (composite FK).
- Court reference only after charge-sheet; verdict only when the court case is DISPOSED (CHECK).
- ID-proof formats per type (Aadhaar, PAN, Voter ID, Passport, DL), Indian mobile numbers and badge numbers validated by CHECK + REGEXP.
- Complaints cannot be deleted and the status history cannot be modified (triggers).

8.3 Constraint inventory

14 PRIMARY KEY, **16** UNIQUE, **26** FOREIGN KEY, **50** CHECK constraints, **17** triggers. CHECK constraints:

Table	Constraint	Expression
app_user	chk_app_officer	((user_role = _utf8mb4\'ADMIN\') or (officer_id is not null))
app_user	chk_app_role	(user_role in (_utf8mb4\'ADMIN\',_utf8mb4\'SHO\',_utf8mb4\'IO\',_utf8mb4\'CUSTODIAN\'))
app_user	chk_app_username	regexp_like(username,_utf8mb4\'^[a-z0-9_.]{3,30}$\')
assignment	chk_assignment_dates	((relieved_at is null) or (relieved_at >= assigned_at))
assignment	chk_assignment_role	(assignment_role in (_utf8mb4\'LEAD\',_utf8mb4\'SUPPORT\',_utf8mb4\'FORENSIC\',_utf8mb4\'SURVEILLANCE\'))
charge	chk_charge_act	(act_name in (_utf8mb4\'BNS\',_utf8mb4\'BNSS\',_utf8mb4\'IT Act\',_utf8mb4\'NDPS Act\',_utf8mb4\'Arms Act\',_u
charge	chk_charge_section	regexp_like(section_code,_utf8mb4\'^[0-9]+[A-Za-z0-9()]*$\')
complaint	chk_complaint_category	(category in...
complaint	chk_complaint_closure	(((status = _utf8mb4\'CLOSED\') and (closure_date is not null) and (closure_type is not null) and (closure_rem and...
complaint	chk_complaint_closuredate	((closure_date is null) or (closure_date >= cast(registered_at as date)))
complaint	chk_complaint_closuretype	((closure_type is null) or (closure_type in (_utf8mb4\'SOLVED\',_utf8mb4\'CONVICTED\',_utf8mb4\'ACQUITTED\',_u
complaint	chk_complaint_incident	(incident_datetime <= registered_at)
complaint	chk_complaint_no	regexp_like(complaint_no,_utf8mb4\'^CMP-[0-9]{4}-[0-9]{5}$\')
complaint	chk_complaint_priority	(priority in (_utf8mb4\'LOW\',_utf8mb4\'MEDIUM\',_utf8mb4\'HIGH\',_utf8mb4\'CRITICAL\'))
complaint	chk_complaint_status	(status in (_utf8mb4\'REGISTERED\',_utf8mb4\'UNDER_INVESTIGATION\',_utf8mb4\'CHARGE_SHEETED\',_utf8mb4\'IN_COU
complaint_accused	chk_accused_status	(accused_status in (_utf8mb4\'SUSPECT\',_utf8mb4\'ARRESTED\',_utf8mb4\'ABSCONDING\',_utf8mb4\'ON_BAIL\',_utf8m
court_reference	chk_court_hearing	((next_hearing_date is null) or (next_hearing_date >= filing_date))
court_reference	chk_court_stage	(case_stage in (_utf8mb4\'FILED\',_utf8mb4\'HEARING\',_utf8mb4\'ARGUMENTS\',_utf8mb4\'JUDGMENT_RESERVED\',_utf
court_reference	chk_court_type	(court_type in (_utf8mb4\'MAGISTRATE\',_utf8mb4\'SESSIONS\',_utf8mb4\'HIGH_COURT\',_utf8mb4\'SPECIAL\'))
court_reference	chk_court_verdict	(verdict in (_utf8mb4\'PENDING\',_utf8mb4\'CONVICTED\',_utf8mb4\'ACQUITTED\',_utf8mb4\'DISMISSED\',_utf8mb4\'S
court_reference	chk_court_verdict_stage	((verdict = _utf8mb4\'PENDING\') = (case_stage <> _utf8mb4\'DISPOSED\'))
custody_log	chk_custody_from	((purpose = _utf8mb4\'INITIAL_SEIZURE\') = (from_officer_id is null))
custody_log	chk_custody_location	(char_length(trim(location)) >= 3)
custody_log	chk_custody_period	((released_at is null) or (released_at > received_at))
custody_log	chk_custody_purpose	(purpose in (_utf8mb4\'INITIAL_SEIZURE\',_utf8mb4\'STORAGE\',_utf8mb4\'FORENSIC_ANALYSIS\',_utf8mb4\'INVESTIGA

custody_log	chk_custody_seal	(seal_intact in (0,1))
custody_log	chk_custody_self	((from_officer_id is null) or (from_officer_id <> to_officer_id))
evidence_item	chk_evidence_category	(category in (_utf8mb4\'WEAPON\',_utf8mb4\'DOCUMENT\',_utf8mb4\'ELECTRONIC\',_utf8mb4\'BIOLOGICAL\',_utf8mb4\'NARCOTIC\',_u
evidence_item	chk_evidence_condition	(item_condition in (_utf8mb4\'INTACT\',_utf8mb4\'DAMAGED\',_utf8mb4\'PARTIAL\',_utf8mb4\'PERISHABLE\'))
evidence_item	chk_evidence_no	regexp_like(evidence_no,_utf8mb4\'^EVD-[0-9]{4}-[0-9]{5}$\')
evidence_item	chk_evidence_qty	(quantity > 0)
investigation_activity	chk_activity_desc	(char_length(trim(description)) >= 5)
investigation_activity	chk_activity_type	(activity_type in (_utf8mb4\'SITE_VISIT\',_utf8mb4\'STATEMENT_RECORDED\',_utf8mb4\'INTERROGATION\',_utf8mb4\'RAID\',_utf8mb4\'AR
officer	chk_officer_active	(is_active in (0,1))
officer	chk_officer_badge	regexp_like(badge_no,_utf8mb4\'^MH[0-9]{2}-[0-9]{4}$\')
officer	chk_officer_joined	(date_joined >= _utf8mb4\'1970-01-01\')
officer	chk_officer_phone	regexp_like(phone,_utf8mb4\'^[6-9][0-9]{9}$\')
officer	chk_officer_rank	(officer_rank in (_utf8mb4\'Constable\',_utf8mb4\'Head Constable\',_utf8mb4\'Assistant Sub-Inspector\',_utf8mb
person	chk_person_dob	((date_of_birth is null) or (date_of_birth >= _utf8mb4\'1900-01-01\'))
person	chk_person_gender	(gender in (_utf8mb4\'M\',_utf8mb4\'F\',_utf8mb4\'O\'))
person	chk_person_idformat	((id_proof_type is null) or ((id_proof_type = _utf8mb4\'AADHAAR\') and regexp_like(id_proof_no,_utf8mb4\'^[2-9 ((id_proof_type =...
person	chk_person_idpair	((id_proof_type is null) = (id_proof_no is null))
person	chk_person_idtype	((id_proof_type is null) or (id_proof_type in (_utf8mb4\'AADHAAR\',_utf8mb4\'PAN\',_utf8mb4\'VOTER_ID\',_utf8m
person	chk_person_phone	((phone is null) or regexp_like(phone,_utf8mb4\'^[6-9][0-9]{9}$\'))
station	chk_station_code	regexp_like(station_code,_utf8mb4\'^PS-[A-Z]{3}$\')
status_history	chk_history_change	((old_status is null) or (old_status <> new_status))
status_history	chk_history_new	(new_status in (_utf8mb4\'REGISTERED\',_utf8mb4\'UNDER_INVESTIGATION\',_utf8mb4\'CHARGE_SHEETED\',_utf8mb4\'IN
status_history	chk_history_old	((old_status is null) or (old_status in (_utf8mb4\'REGISTERED\',_utf8mb4\'UNDER_INVESTIGATION\',_utf8mb4\'CHAR
witness	chk_witness_protected	(is_protected in (0,1))
witness	chk_witness_statement	(char_length(trim(statement)) >= 10)

9. Triggers, procedures and functions

Trigger	Timing	Table	Purpose
trg_assignment_bi	BEFORE INSERT	assignment	Active officer only, case not closed, not before registration
trg_charge_bi	BEFORE INSERT	charge	Charges only while under investigation or later
trg_complaint_bi	BEFORE INSERT	complaint	Generate CMP number, normalise registration time, force initial REGISTERED status
trg_complaint_bu	BEFORE UPDATE	complaint	Status workflow + preconditions; complaint number immutable; closed case not editable
trg_complaint_bd	BEFORE DELETE	complaint	Forbid deleting complaints
trg_complaint_ai	AFTER INSERT	complaint	Write the first status-history row
trg_complaint_au	AFTER UPDATE	complaint	Audit trail: write status_history on every status change (acting officer from @current_officer_id)
trg_accused_bi	BEFORE INSERT	complaint_accused	Case open; complainant / witness cannot be accused
trg_court_bi	BEFORE INSERT	court_reference	Court reference only after charge-sheet
trg_custody_bi	BEFORE INSERT	custody_log	Complete-transfer, no-gap, no-overlap and final-disposition checks
trg_custody_bu	BEFORE UPDATE	custody_log	Append-only: only release time can be set, once
trg_custody_bd	BEFORE DELETE	custody_log	Forbid deleting custody records
trg_evidence_bi	BEFORE INSERT	evidence_item	Generate EVD number; case open; collected after incident; active collector
trg_activity_bi	BEFORE INSERT	investigation_activity	Case open, time valid, officer assigned at activity time
trg_history_bu	BEFORE UPDATE	status_history	Audit log cannot be modified
trg_history_bd	BEFORE DELETE	status_history	Audit log cannot be deleted
trg_witness_bi	BEFORE INSERT	witness	Case open; accused cannot be witness

Routine	Type	Purpose
fn_case_age_days	FUNCTION	Age of a case in days
fn_current_holder	FUNCTION	Current custodian of an evidence item
sp_close_complaint	PROCEDURE	Atomic closure: close case and relieve all active officers
sp_transfer_custody	PROCEDURE	Atomic custody transfer (SELECT … FOR UPDATE, release + insert, ROLLBACK on error)

```
-- excerpt: no overlapping custody (trg_custody_bi)
SELECT COUNT(*) INTO v_overlap FROM custody_log
 WHERE evidence_id = NEW.evidence_id
   AND NEW.received_at < COALESCE(released_at, '9999-12-31 23:59:59')
   AND COALESCE(NEW.released_at, '9999-12-31 23:59:59') > received_at;
IF v_overlap > 0 THEN
   SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Overlapping custody period for this evidence item.';
END IF;
```

10. Views and SQL queries

View	Report	Techniques
v_pending_cases	Pending cases	JOIN, LEFT JOIN, correlated sub-queries, DATEDIFF
v_officer_load	Officer load	LEFT JOIN, COUNT(DISTINCT CASE ...), GROUP BY, sub-queries
v_evidence_location	Evidence location	JOIN on open custody row, derived status via CASE
v_custody_history	Custody history	Self-referencing officer joins, ROW_NUMBER() window, TIMESTAMPDIFF
v_case_witnesses	Witnesses	Multi-table JOIN
v_case_timeline	Case timeline	UNION ALL of 8 event sources
v_closed_cases	Closures	JOIN, scalar sub-queries, DATEDIFF
v_case_summary	Case list (application)	JOIN + LEFT JOIN + sub-queries
v_upcoming_hearings	Upcoming hearings	JOIN, date filter

`sql/05_report_queries.sql` contains 25 demonstration queries: the 7 required reports, multi-table INNER/LEFT joins, cross-station custody hand-overs, GROUP BY / HAVING / ROLLUP, conditional aggregation (pivot), nested and correlated sub-queries, EXISTS / NOT EXISTS / IN, derived tables, window functions (RANK, ROW_NUMBER, LAG, LEAD), a CTE and a stored function call.

```
-- Q22 Time spent in each status (LEAD window function over the audit trail)
WITH status_periods AS (
    SELECT complaint_id, new_status, changed_at,
           LEAD(changed_at) OVER (PARTITION BY complaint_id ORDER BY changed_at) AS next_change
      FROM status_history)
SELECT c.complaint_no, sp.new_status, sp.changed_at,
       DATEDIFF(COALESCE(sp.next_change, NOW()), sp.changed_at) AS days_in_status
  FROM status_periods sp JOIN complaint c ON c.complaint_id = sp.complaint_id;

-- Q17 Open cases with no activity in the last 30 days (NOT EXISTS)
SELECT c.complaint_no, c.title, c.status FROM complaint c
 WHERE c.status <> 'CLOSED'
   AND NOT EXISTS (SELECT 1 FROM investigation_activity a
                   WHERE a.complaint_id = c.complaint_id
                     AND a.activity_time >= CURDATE() - INTERVAL 30 DAY);
```

11. Indexing strategy

InnoDB automatically indexes primary keys, unique keys and foreign keys. Additional secondary indexes were created for the most frequent search, filter, sort and join paths:

Index	Columns	Used by
idx_assignment_officer	assignment(officer_id,relieved_at)	Officer load (active assignments)
ft_complaint_text	complaint(title,description)	Full-text search on title and description
idx_complaint_category	complaint(category)	Category filter, analytics
idx_complaint_incident	complaint(incident_datetime)	Date-range queries
idx_complaint_priority	complaint(priority)	Priority filter / sort
idx_complaint_status_reg	complaint(status,registered_at)	Pending/closed lists sorted by date, dashboard counts
idx_court_next_hearing	court_reference(next_hearing_date)	Upcoming hearings
idx_custody_evidence_time	custody_log(evidence_id,received_at)	Chain of custody ordering, overlap check in trigger
idx_custody_holder	custody_log(to_officer_id,released_at)	Evidence held by an officer
idx_evidence_category	evidence_item(category)	Evidence filter
idx_activity_case_time	investigation_activity(complaint_id,activity_time)	Case diary, last activity, stale cases
idx_officer_name	officer(full_name)	Officer search
idx_person_name	person(full_name)	Person search / picker
idx_person_phone	person(phone)	Search by phone
idx_history_case_time	status_history(complaint_id,changed_at)	Status audit / timeline

12. Transactions and concurrency

The application uses an explicit `START TRANSACTION … COMMIT` block (`db.transaction()`, isolation READ COMMITTED); any exception triggers `ROLLBACK`, so a multi-step operation is all-or-nothing:

Operation	Statements in one transaction
Register complaint	INSERT person (if new) → INSERT complaint (+ trigger history) → optional INSERT assignment + UPDATE status
Register evidence	INSERT evidence_item → INSERT custody (INITIAL_SEIZURE) → optional release + INSERT custody (STORAGE)
Custody transfer	**SELECT … FOR UPDATE** on the open custody row → UPDATE released_at → INSERT new period
Status change	SELECT status FOR UPDATE (optimistic check) → UPDATE complaint (+ audit trigger) → relieve officers on closure
Replace lead / role change	Lock current lead → relieve → INSERT new assignment
Add witness / accused with new person	INSERT person → INSERT witness / complaint_accused

The row lock taken with `FOR UPDATE` serialises two custodians trying to transfer the same exhibit at the same moment — the second waits, then fails the no-overlap check instead of creating two current holders. Test **TC-10** proves rollback: a valid new person is *not* saved when the complaint insert fails. Stored procedures `sp_transfer_custody` and `sp_close_complaint` show the same pattern in pure SQL with an EXIT HANDLER that rolls back.

13. Application design

13.1 Architecture

```
Browser (HTML/CSS/JS)  ──HTTP──▶  Flask routes (blueprints)      complaints, evidence, persons, officers, reports, admin, auth
                                      │  form validation (validators.py) · RBAC (auth.requires) · CSRF
                                      ▼
                                  Services (business logic + transactions)
                                      │  parameterised SQL only
                                      ▼
                                  db.py  — connection pool, transaction(), error translation
                                      ▼
                                  MySQL 8: tables · constraints · triggers · views · procedures
```

13.2 Modules

Module	Features
Dashboard	KPIs, status donut, registrations vs closures chart, officer workload, crime categories, items needing attention, live activity feed
Complaints	Search/filter/sort/paginate; register (existing or new complainant, optional lead); case page with status stepper and tabs: overview, officers, investigation diary, evidence, accused & witnesses, charges & court, timeline, status audit; printable case file
Evidence & custody	Malkhana register with derived status; register evidence (+ deposit); visual chain of custody; atomic transfer dialog; printable custody sheet
Persons / Officers	Full CRUD, involvement across cases, workload profile, activation/deactivation
Reports	12 parameterised reports with charts, quick filter, print and CSV export
Integrity lab	Runs 26 invalid SQL scenarios against MySQL (always rolled back) and shows the database error for each rule
Database explorer	Tables, columns, constraints, relationships, triggers, routines and indexes from information_schema

13.3 Validation and error handling

- **Two layers:** application validation (required fields, lengths, formats, dates not in the future, allowed choices) gives field-level messages; the database re-checks everything as the final authority.
- MySQL errors are translated into readable messages: 1062 duplicate → which key, 3819 CHECK → rule description, 1451/1452 FK, 1644 trigger SIGNAL text, 1048 NOT NULL. The form is re-displayed with the user's input and the dialog re-opened.
- Custom 400/403/404/500/503 pages; unexpected errors are logged to `logs/app.log`; a database-down page explains how to fix the connection.

13.4 Security

- No login page (not required by the brief): the acting user is chosen from the sidebar; session cookies are HttpOnly + SameSite. `app_user` keeps PBKDF2 password hashes so a login screen can be re-enabled for real deployment.
- Role-based access checked on every route; CSRF token on every POST form; all SQL parameterised (no string concatenation of user input).
- ID-proof numbers masked in lists; protected-witness names hidden in the printable case file.

14. Testing

14.1 Strategy

- **Automated functional tests** (`tests/test_app.py`, Python unittest): drive the real application through HTTP against a separate database `police_evidence_test` rebuilt from the SQL scripts on each run.
- **Database negative tests** (`sql/06_constraint_tests.sql`): 28 invalid statements, each must fail.
- **Integrity Lab**: 26 live scenarios proving every rule is enforced by MySQL itself.
- **Manual UI tests** on desktop and mobile widths, light and dark themes (see output screens).

14.2 Automated test results — 43 / 43 passed (4.9 s)

ID	Module	Scenario	Expected result	Result
TC-01	Acting user	Open the app with a fresh browser	Dashboard opens directly; no login page	PASS
TC-02	Acting user	Check the default acting user	Session uses default_user from config.ini (admin)	PASS
TC-03	Acting user	Switch to anjali, then to an unknown user	Role/name change; unknown user rejected	PASS
TC-04	Security (RBAC)	Custodian opens Register complaint / Add officer	HTTP 403; evidence registration allowed	PASS
TC-05	Security	POST the user switcher without CSRF token	HTTP 400 - request rejected	PASS
TC-06	Complaint registration	Register complaint with a NEW complainant	CMP-YYYY-NNNNN issued, status REGISTERED, person + history row created	PASS
TC-07	Complaint registration	Register complaint and assign lead officer at once	Status UNDER_INVESTIGATION, LEAD assignment, 2 history rows	PASS
TC-08	Input validation	Empty title, short description, invalid category, future incident time	HTTP 422 with field errors; nothing inserted	PASS
TC-09	Input validation	Phone '12345', PAN '1234'	Field errors for phone and PAN format	PASS
TC-10	Transactions	Valid new person + complaint with non-existent station	FK error; whole transaction rolled back - person NOT saved	PASS
TC-11	Officer assignment	Assign LEAD and FORENSIC officers	2 active assignments	PASS
TC-12	Officer assignment	Assign a second LEAD without 'replace'	Rejected: case already has an active LEAD	PASS
TC-13	Transactions	Replace current lead	Old lead relieved + new lead inserted atomically	PASS
TC-14	Officer assignment	Assign retired (inactive) officer	Rejected by trigger trg_assignment_bi	PASS
TC-15	Status update / activity	Move to UNDER_INVESTIGATION, log a site visit	Activity saved; audit row records acting officer & remarks	PASS
TC-16	Activity logging	Officer not on the case logs an activity	Rejected by trigger trg_activity_bi	PASS
TC-17	Input validation	Activity dated tomorrow	'cannot be in the future'	PASS
TC-18	Evidence registration	Register evidence and deposit with custodian	EVD number issued; 2 custody periods with no gap; custodian is holder	PASS
TC-19	Evidence registration	Second item with same seal number	Rejected: seal number already used (UNIQUE)	PASS
TC-20	Custody transfer	Transfer custodian -> forensic officer	Previous period closed at transfer time; new period opened; status IN_LAB	PASS
TC-21	Custody transfer	Transfer dated before current custody began	Rejected; chain unchanged	PASS
TC-22	Custody transfer	Seal not intact and no remarks	Validation error - remarks mandatory	PASS
TC-23	Custody transfer	Transfer an item already returned to owner	Rejected - custody chain is final	PASS
TC-24	Integrity (DB level)	Run all 26 Integrity Lab scenarios with raw SQL	Every invalid statement rejected by MySQL	PASS
TC-25	Witness entry	Record protected witness statement (new person)	Witness row + person created; protected flag set	PASS
TC-26	Accused	Add an existing witness as accused in same case	Rejected by trigger trg_accused_bi	PASS
TC-27	Charges	Add accused, frame BNS 303(2) charge	Charge saved	PASS
TC-28	Status update	Charge-sheet a case with no charges / with charges	First rejected, second accepted	PASS
TC-29	Status update	CHARGE_SHEETED -> REGISTERED	Rejected: invalid transition	PASS
TC-30	Court reference	Add court reference, move to IN_COURT	Both succeed	PASS
TC-31	Court reference	Verdict CONVICTED while stage HEARING	Validation error	PASS
TC-32	Closure	Close case without closure details	Validation error; status unchanged	PASS
TC-33	Security (RBAC)	Investigating Officer tries to close case	HTTP 403	PASS
TC-34	Closure / transactions	Close with all closure details	CLOSED; all officers relieved in same transaction; appears in closures view	PASS
TC-35	Closure	Log activity on a closed case	Rejected	PASS
TC-36	Status update	Re-open closed case	UNDER_INVESTIGATION, closure fields cleared, full audit trail kept	PASS

TC-37	CRUD	Create, update and delete an unreferenced person	All three succeed; ID proof upper-cased	**PASS**
TC-38	CRUD / FK	Delete a complainant	Friendly 'referenced by other records' message (FK 1451)	**PASS**
TC-39	CRUD	Bad badge format, valid officer, deactivate busy officer	Error, success, refusal	**PASS**
TC-40	Search	Exact complaint number, name search, filtered list	Redirect to case; matching results	**PASS**
TC-41	Reports	Open every report and export CSV	All 200 OK, CSV mimetype	**PASS**
TC-42	Reports	Pending cases view vs closed cases	No overlap	**PASS**
TC-43	Integrity (DB level)	LAG() audit over the whole custody log	0 broken links	**PASS**

14.3 Database-level integrity results

26 / 26 invalid operations were rejected by MySQL.

Rule	Scenario	Enforced by	MySQL response
R1	Duplicate complaint number	`UNIQUE uq_complaint_no`	ERROR 1062: Duplicate entry 'CMP-2026-00001' for key 'complaint.uq_complaint_no'
R1	Duplicate evidence number	`UNIQUE uq_evidence_no`	ERROR 1062: Duplicate entry 'EVD-2026-00001' for key 'evidence_item.uq_evidence_no'
R1	Malformed complaint number	`CHECK chk_complaint_no (REGEXP)`	ERROR 3819: Check constraint 'chk_complaint_no' is violated.
R2	Transfer without handing officer	`Trigger trg_custody_bi + CHECK chk_custody_from`	ERROR 1644: Incomplete custody transfer: the handing officer is not the current custodian.
R2	Handing officer is not the current custodian	`Trigger trg_custody_bi`	ERROR 1644: Incomplete custody transfer: the handing officer is not the current custodian.
R2	New custody while current holder has not released	`Trigger trg_custody_bi`	ERROR 1644: Incomplete custody transfer: the current custodian has not released the item.
R2	Transfer after evidence returned to owner	`Trigger trg_custody_bi`	ERROR 1644: Evidence has already been returned/disposed - the custody chain is final.
R3	Unknown status value	`Trigger trg_complaint_bu + CHECK chk_complaint_status`	ERROR 1644: Invalid status change: REGISTERED -> PENDING.
R3	Illegal workflow jump	`Trigger trg_complaint_bu`	ERROR 1644: Invalid status change: REGISTERED -> IN_COURT.
R3	Investigate without an officer	`Trigger trg_complaint_bu`	ERROR 1644: Assign at least one officer before starting the investigation.
R3	Charge-sheet without charges	`Trigger trg_complaint_bu`	ERROR 1644: A case cannot be charge-sheeted without at least one charge.
R4	Two current custodians	`Trigger trg_custody_bi + UNIQUE uq_custody_one_open`	ERROR 1644: Incomplete custody transfer: the current custodian has not released the item.
R4	Back-dated overlapping period	`Trigger trg_custody_bi (interval overlap check)`	ERROR 1644: Custody gap/overlap: new custody must start exactly when the previous custody ended.
R4	Tamper with a closed custody record	`Trigger trg_custody_bu (append-only log)`	ERROR 1644: Closed custody records are immutable.
R4	Delete chain-of-custody records	`Trigger trg_custody_bd`	ERROR 1644: Chain-of-custody records cannot be deleted.
R5	Close without closure details	`CHECK chk_complaint_closure`	ERROR 3819: Check constraint 'chk_complaint_closure' is violated.
R5	Closure details on an open case	`CHECK chk_complaint_closure`	ERROR 3819: Check constraint 'chk_complaint_closure' is violated.
R5	Closure date before registration	`CHECK chk_complaint_closuredate`	ERROR 3819: Check constraint 'chk_complaint_closuredate' is violated.
R6	Activity by an unassigned officer	`Trigger trg_activity_bi`	ERROR 1644: Only an officer assigned to this case (at that time) can log an activity.
R6	Assign an inactive officer	`Trigger trg_assignment_bi`	ERROR 1644: Inactive officers cannot be assigned to a case.
R6	Two active LEAD officers	`UNIQUE uq_assignment_one_lead (generated column)`	ERROR 1062: Duplicate entry '5-1' for key 'assignment.uq_assignment_one_lead'
R6	Charge a person who is not accused	`Composite FOREIGN KEY fk_charge_accused`	ERROR 1452: Cannot add or update a child row: a foreign key constraint fails (`police_evidence_db`.`charge`,...
R6	Invalid Aadhaar number	`CHECK chk_person_idformat`	ERROR 3819: Check constraint 'chk_person_idformat' is violated.
R6	Delete a complaint	`Trigger trg_complaint_bd`	ERROR 1644: Complaints are permanent records and cannot be deleted. Close the case instead.
R6	Tamper with the status audit trail	`Trigger trg_history_bu`	ERROR 1644: Status history is an audit log and cannot be modified.
R6	Delete a person referenced by a complaint	`FOREIGN KEY fk_complaint_complainant`	ERROR 1451: Cannot delete or update a parent row: a foreign key constraint fails (`police_evidence_db`.`complaint`,...

15. Output screens

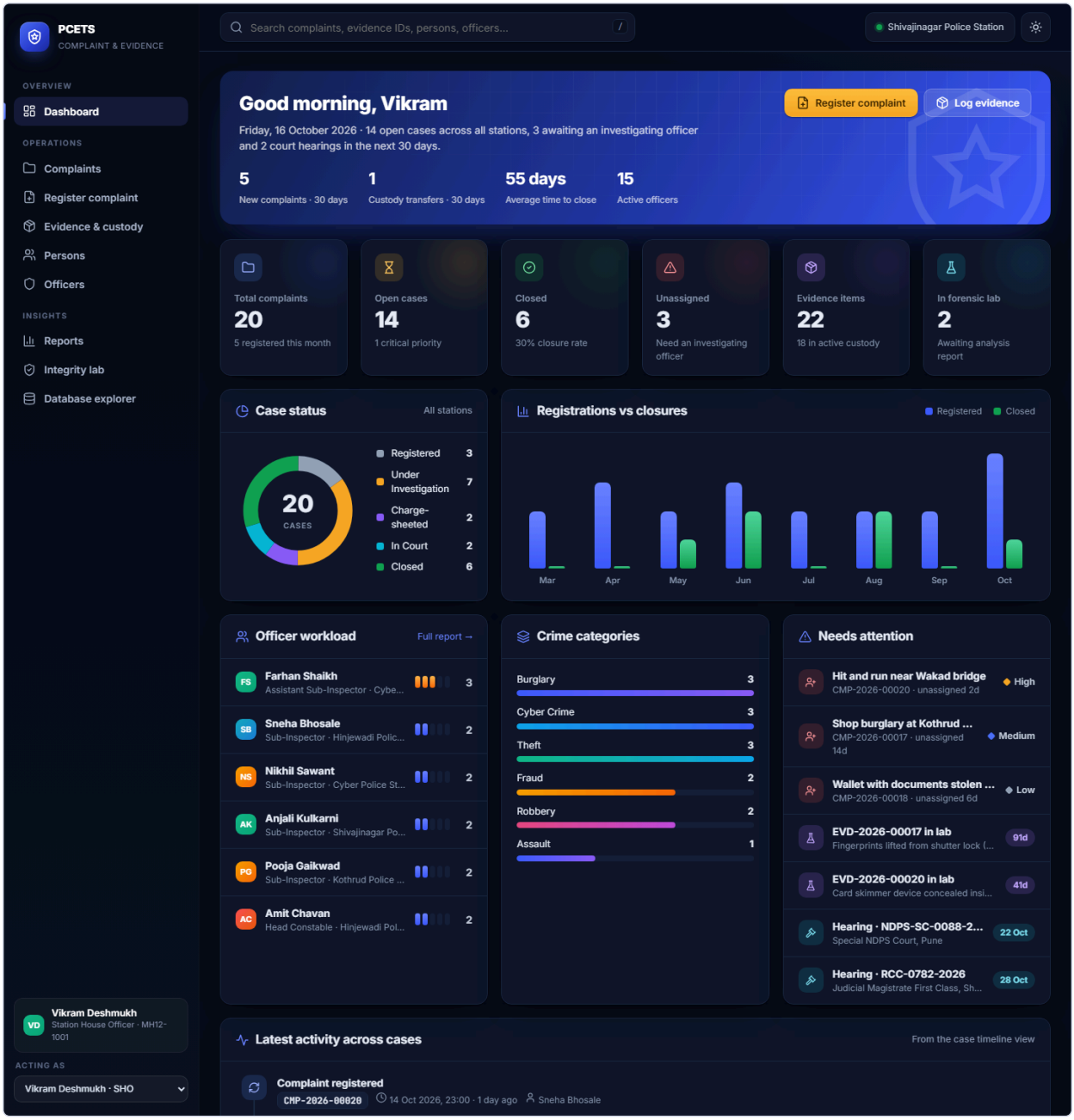


Figure 1. Dashboard — KPIs, status donut, trend chart, workload, attention list

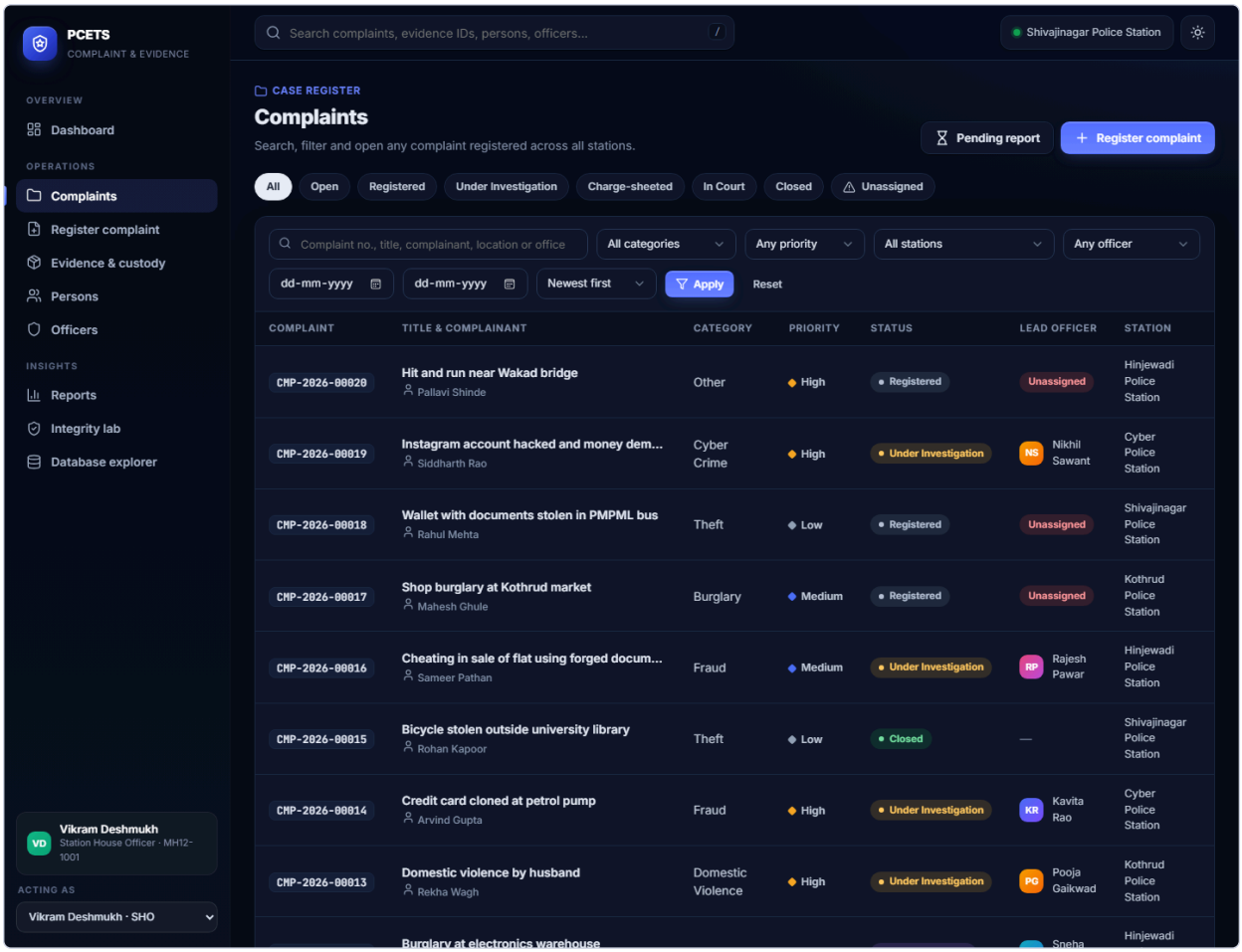


Figure 2. Complaint register with search, filters, sorting and pagination

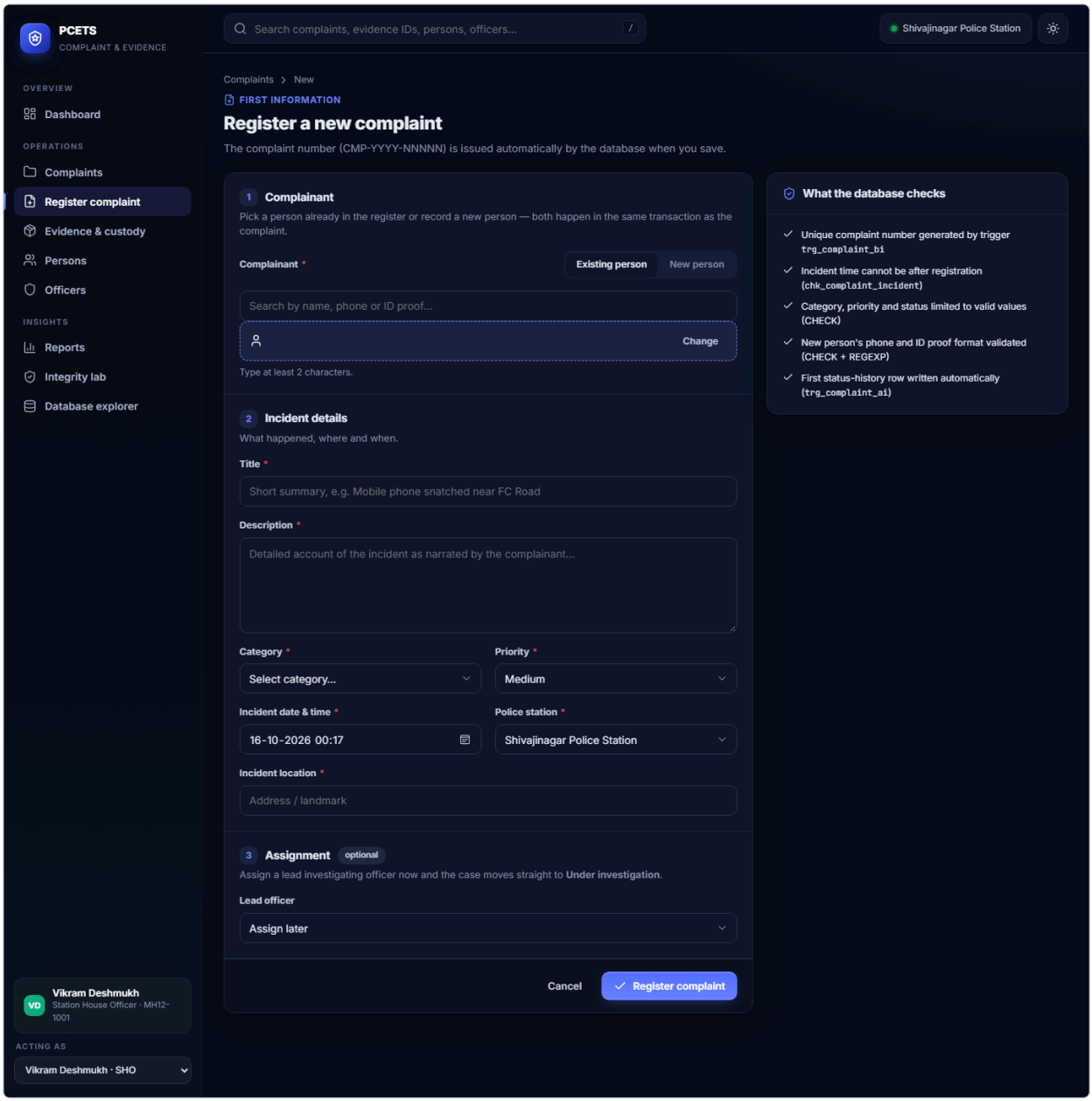


Figure 3. Complaint registration (existing / new complainant, optional lead)

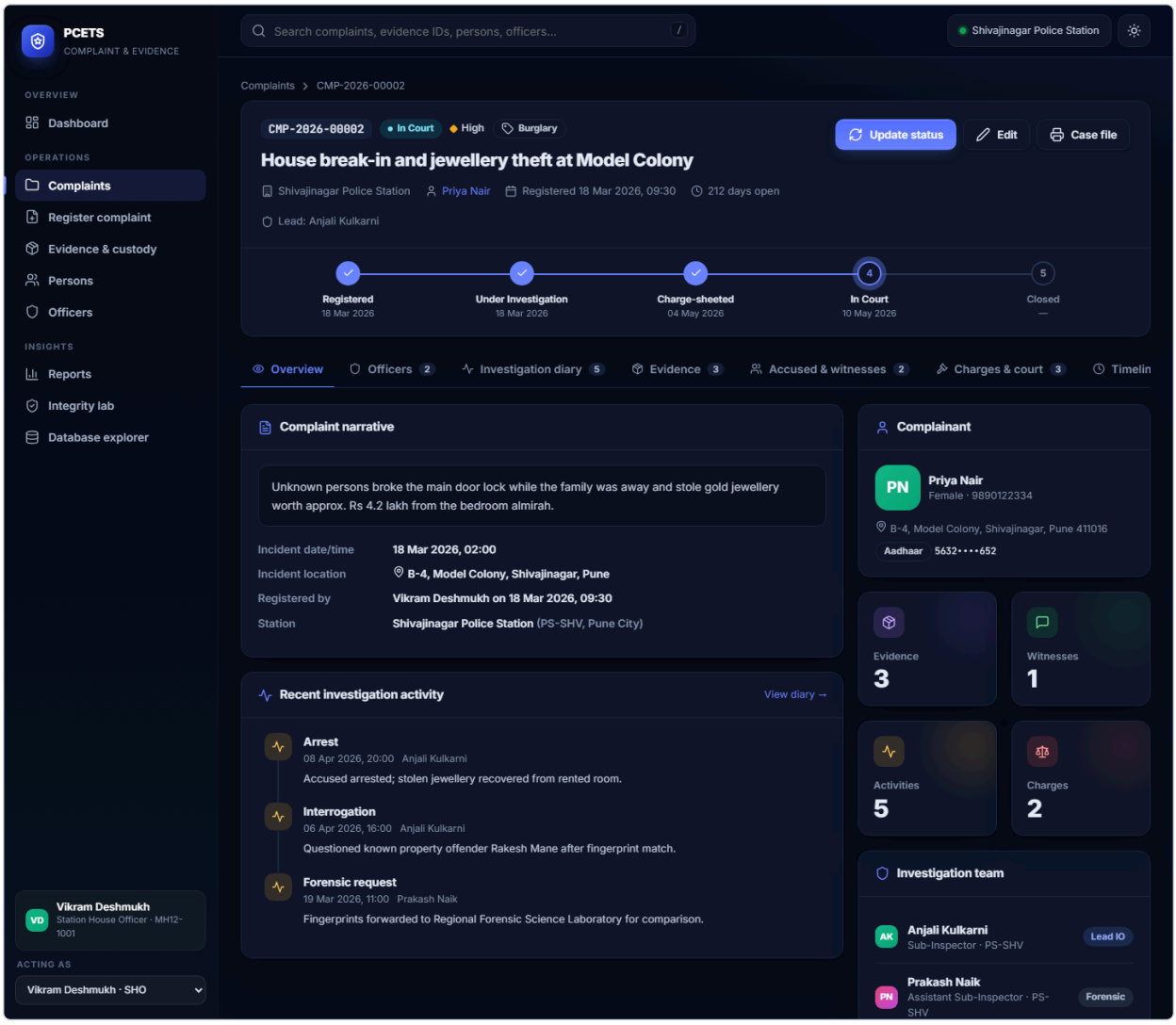


Figure 4. Case page — status stepper and overview

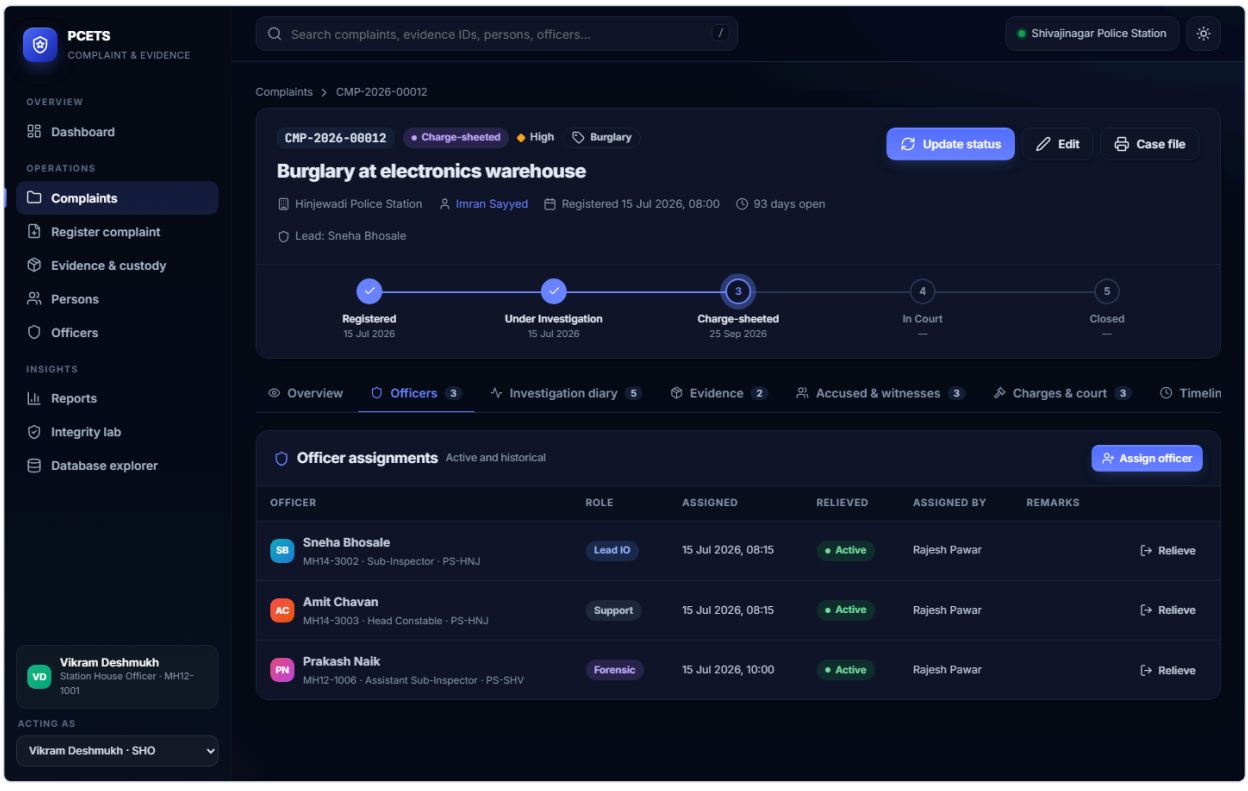


Figure 5. Officer assignments with history

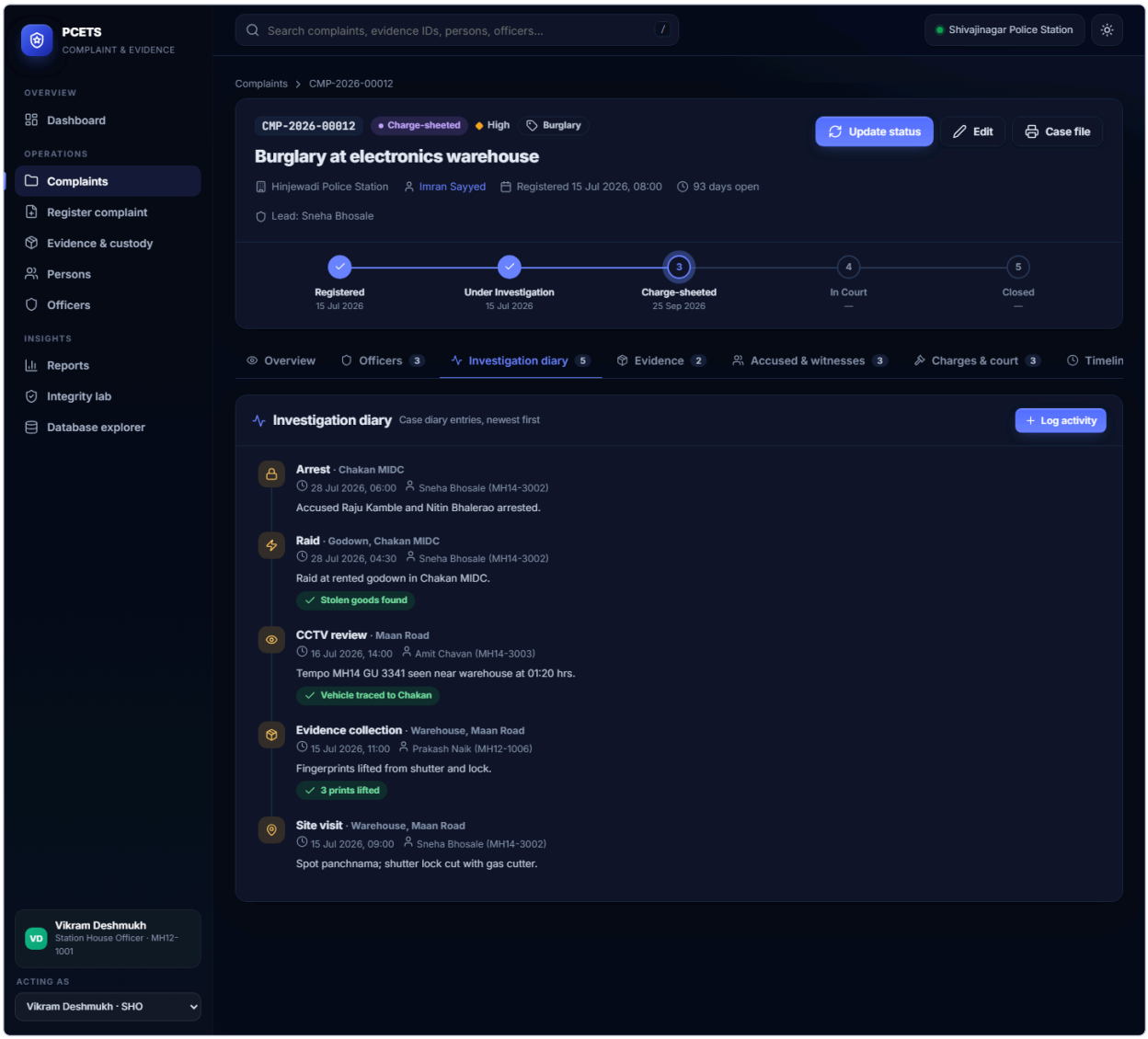


Figure 6. Investigation diary (activity log)

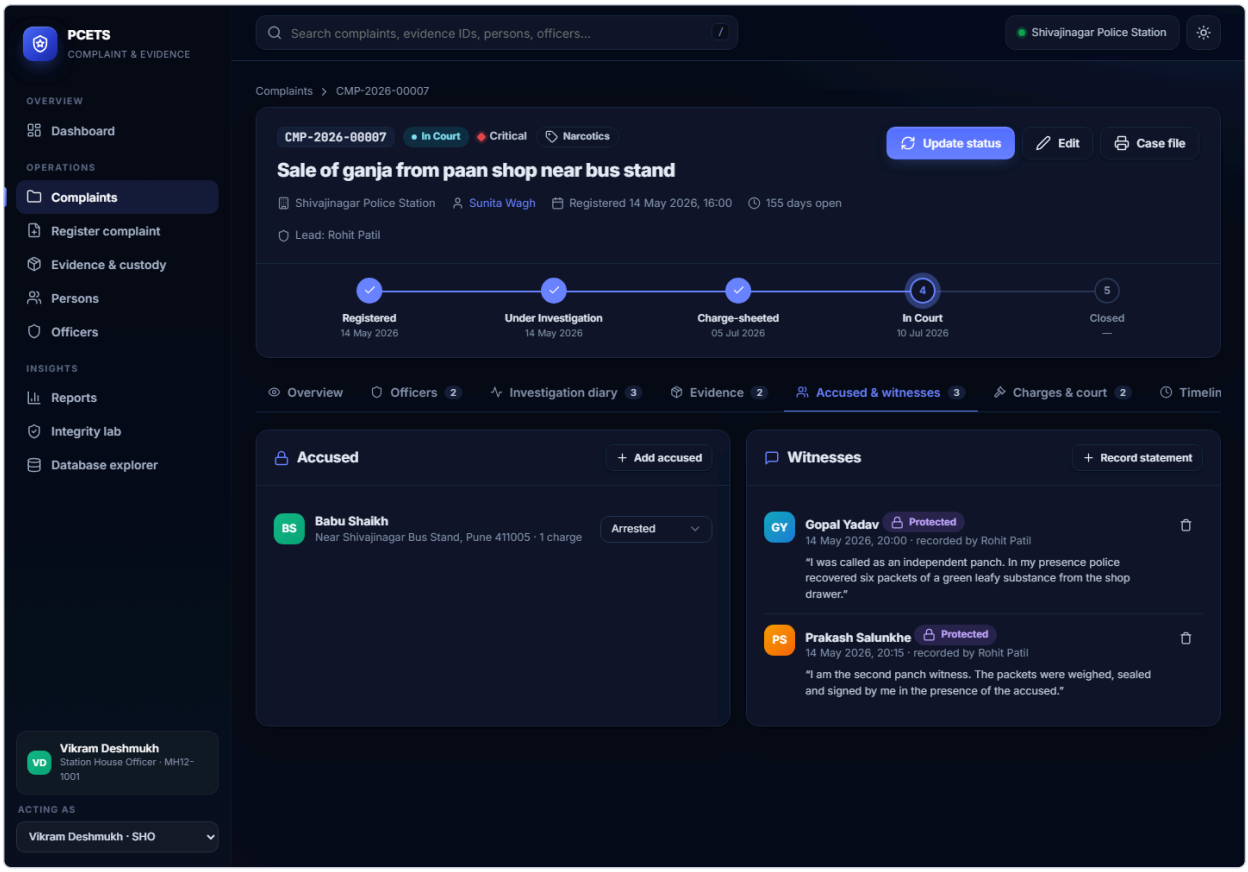


Figure 7. Accused and witnesses (protected witnesses)

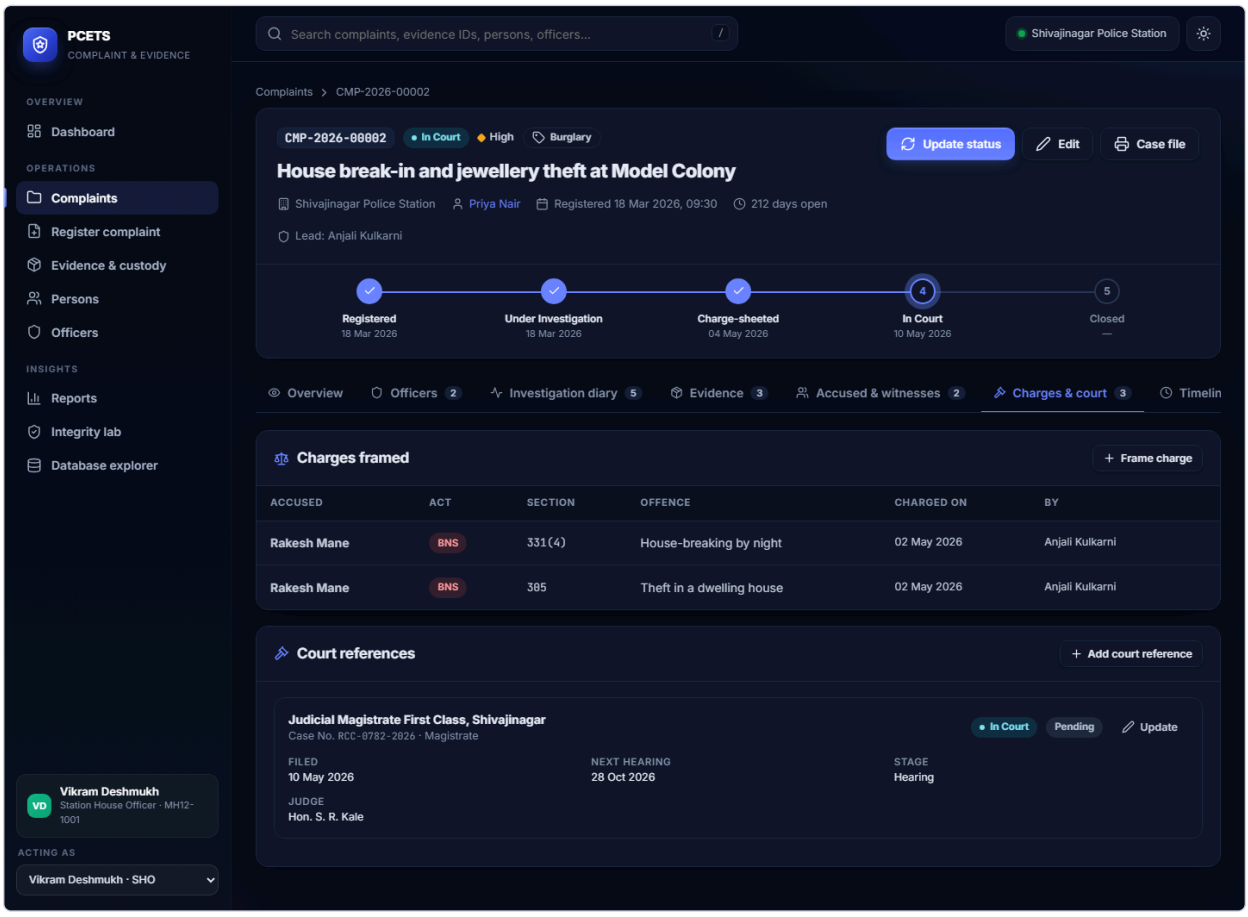


Figure 8. Charges and court references

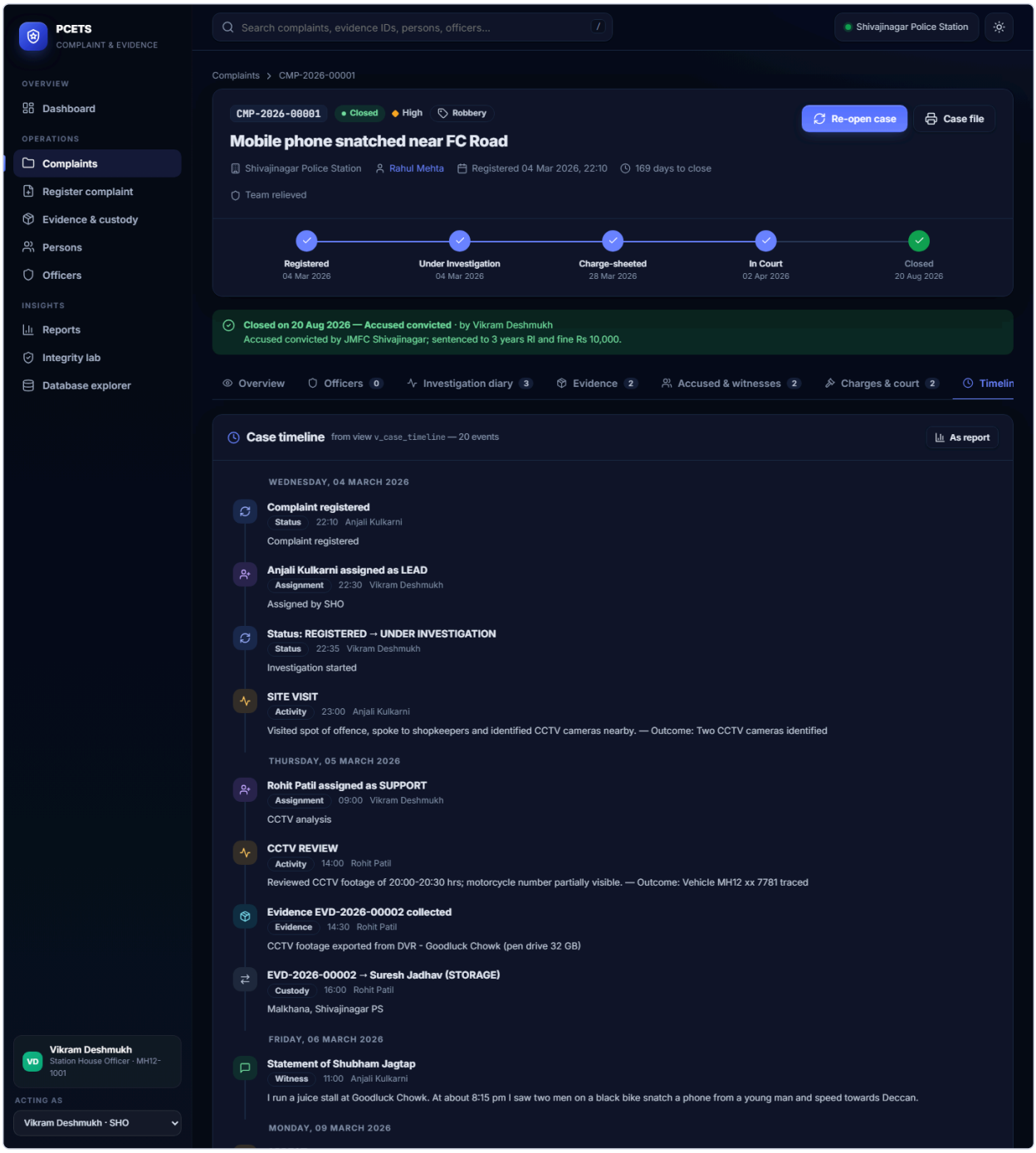


Figure 9. Complete case timeline (v_case_timeline)

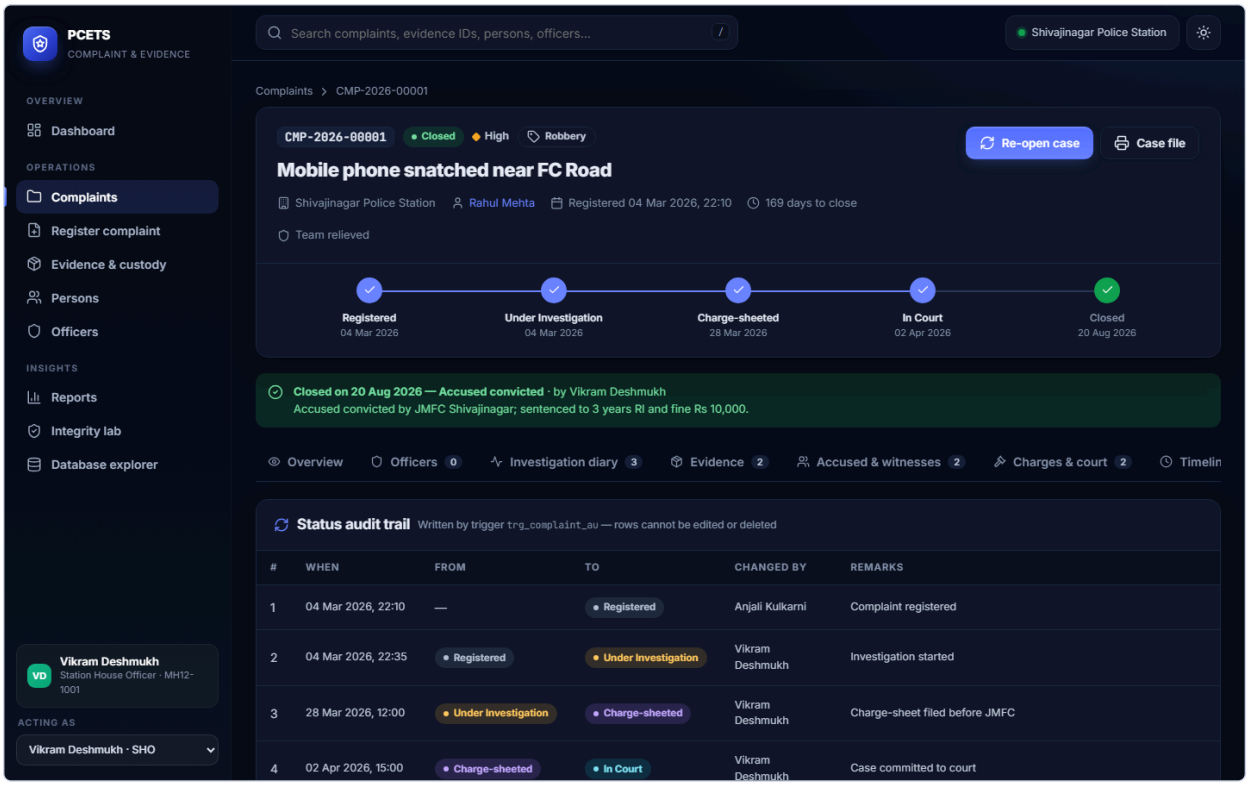


Figure 10. Status audit trail written by trigger

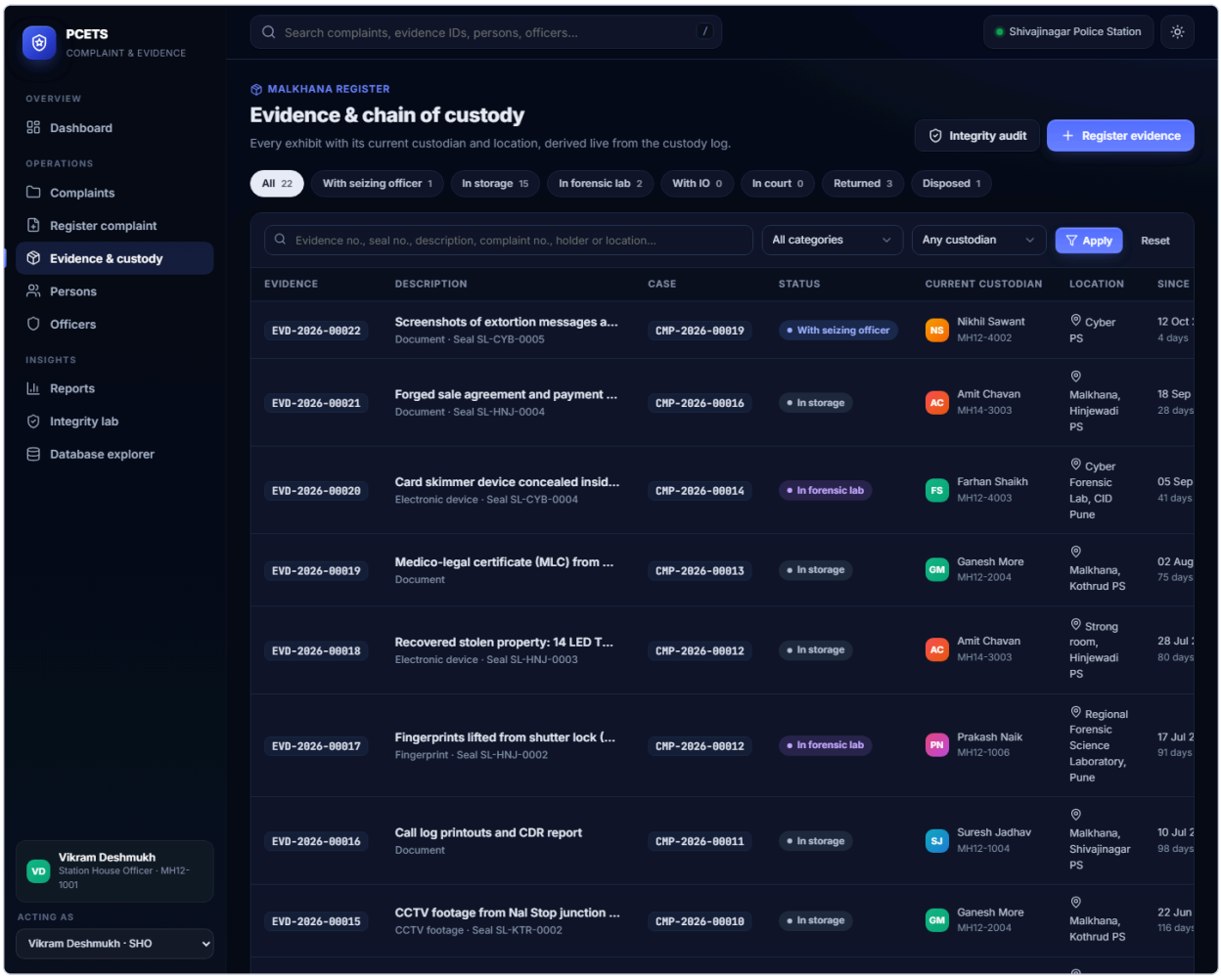


Figure 11. Malkhana register with derived status

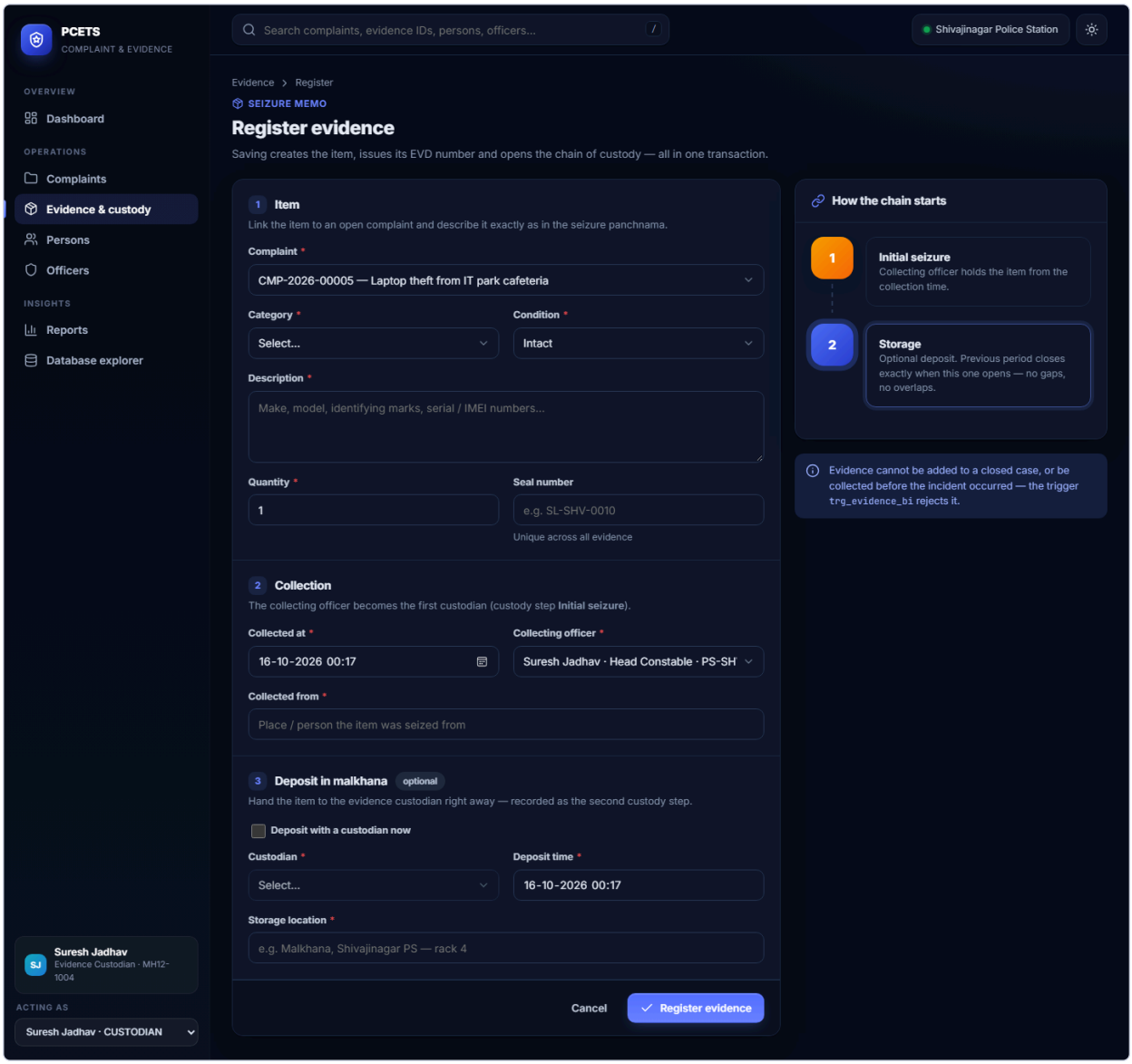


Figure 12. Evidence registration with optional deposit

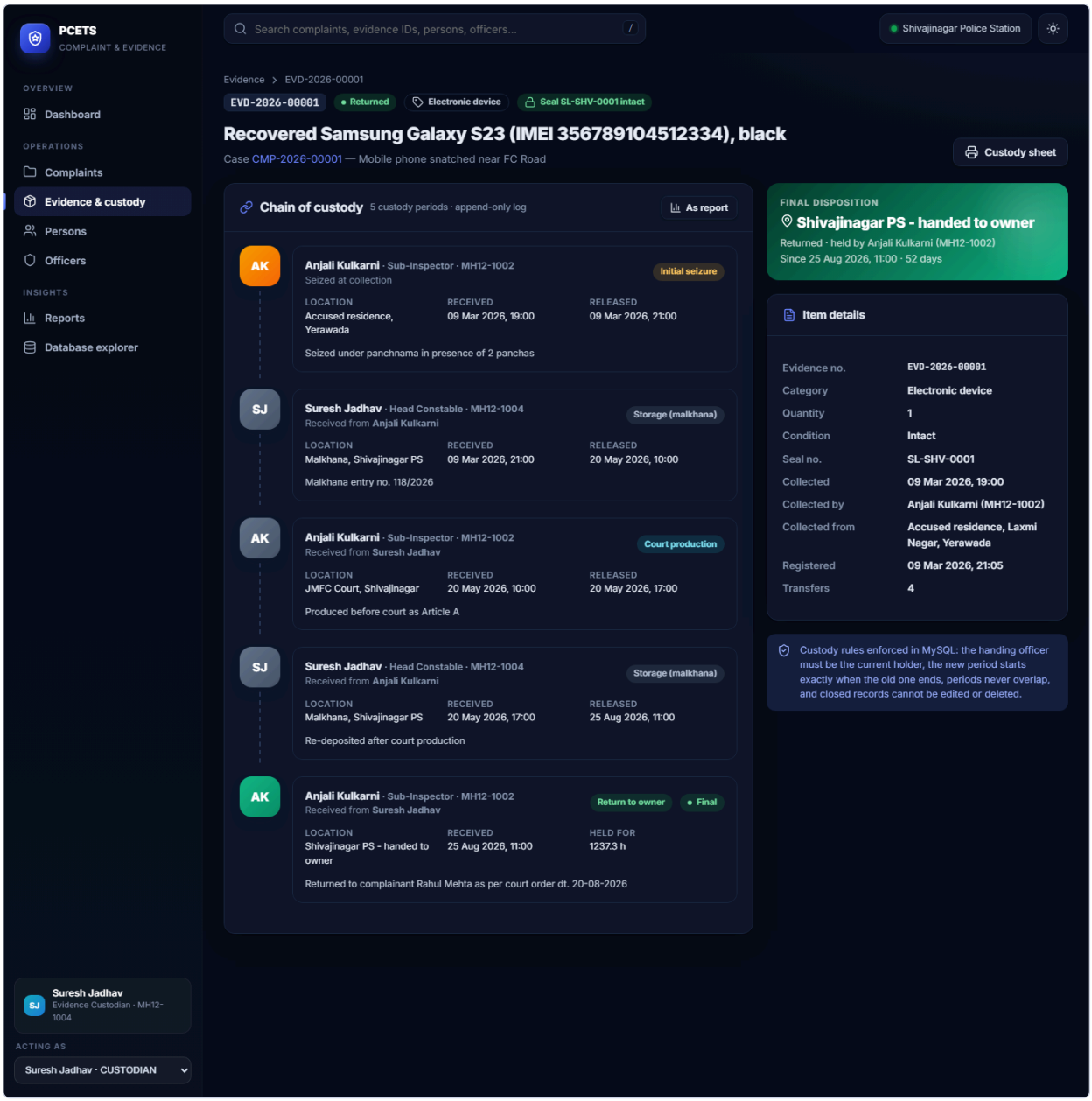


Figure 13. Chain of custody — item returned to owner (final)

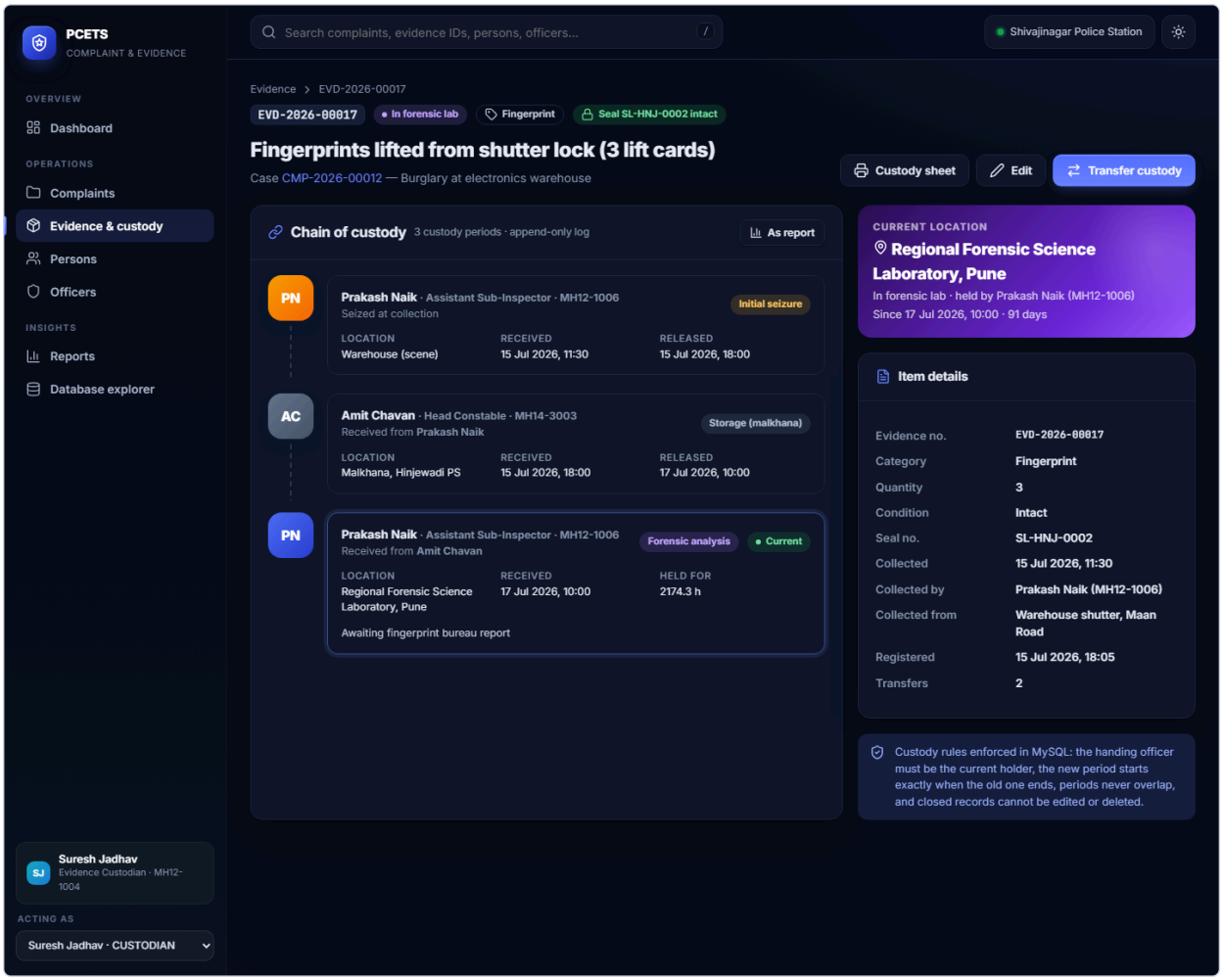


Figure 14. Evidence currently in the forensic lab

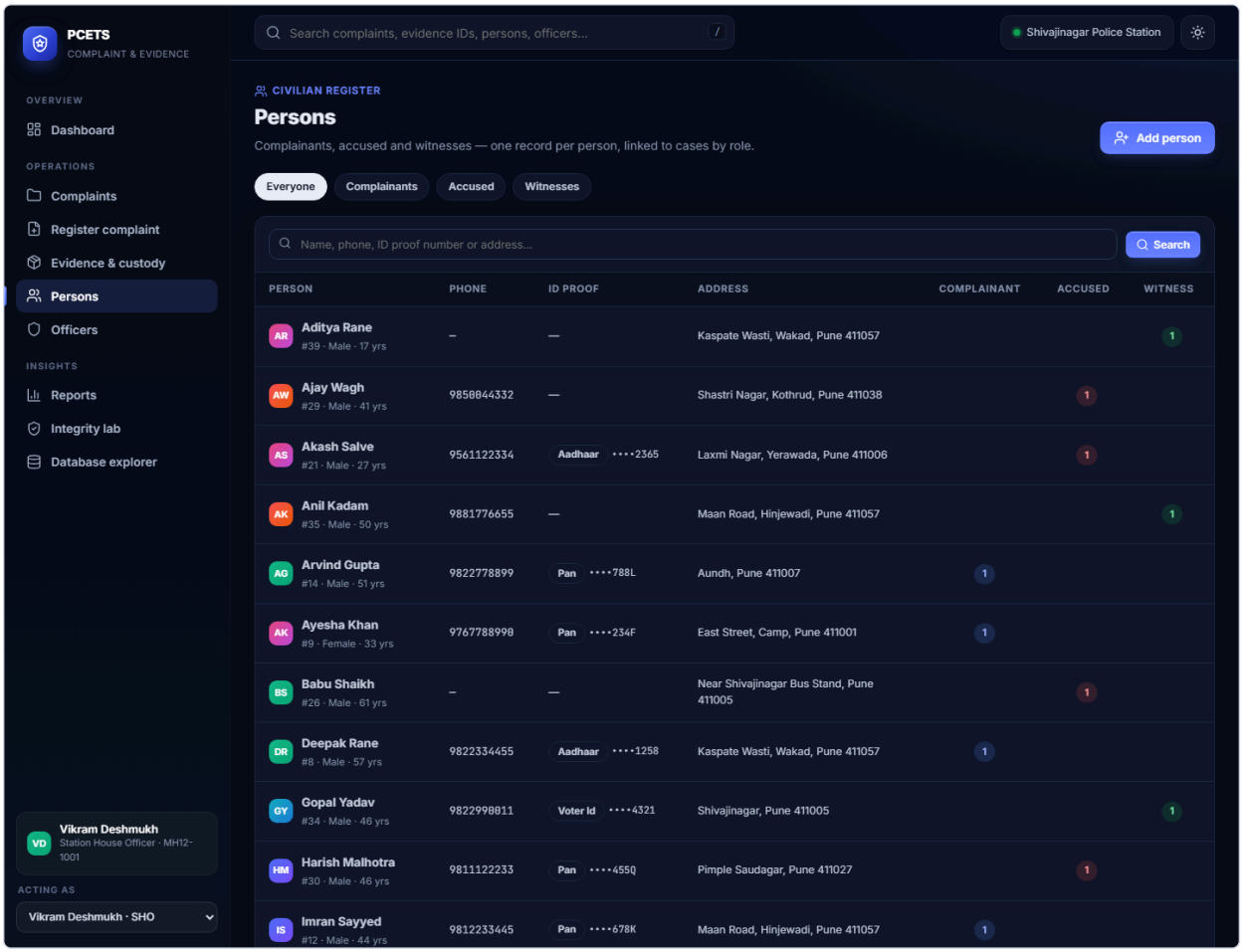


Figure 15. Person register with roles

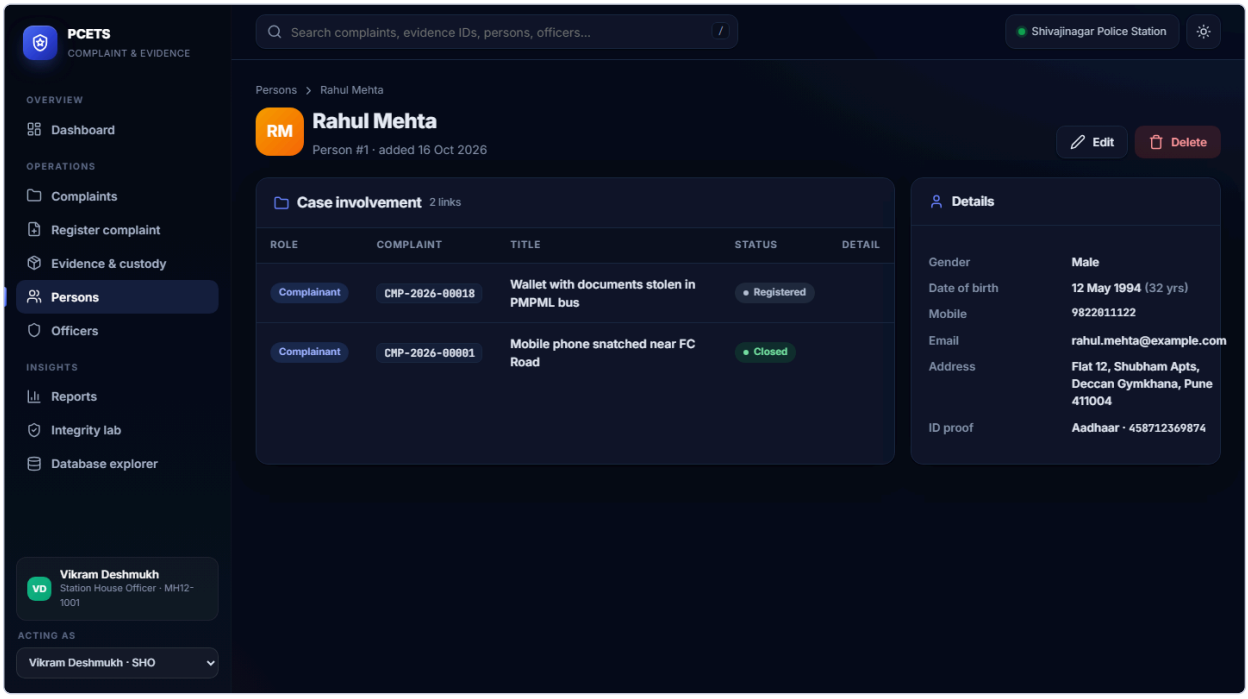


Figure 16. Person involvement across cases

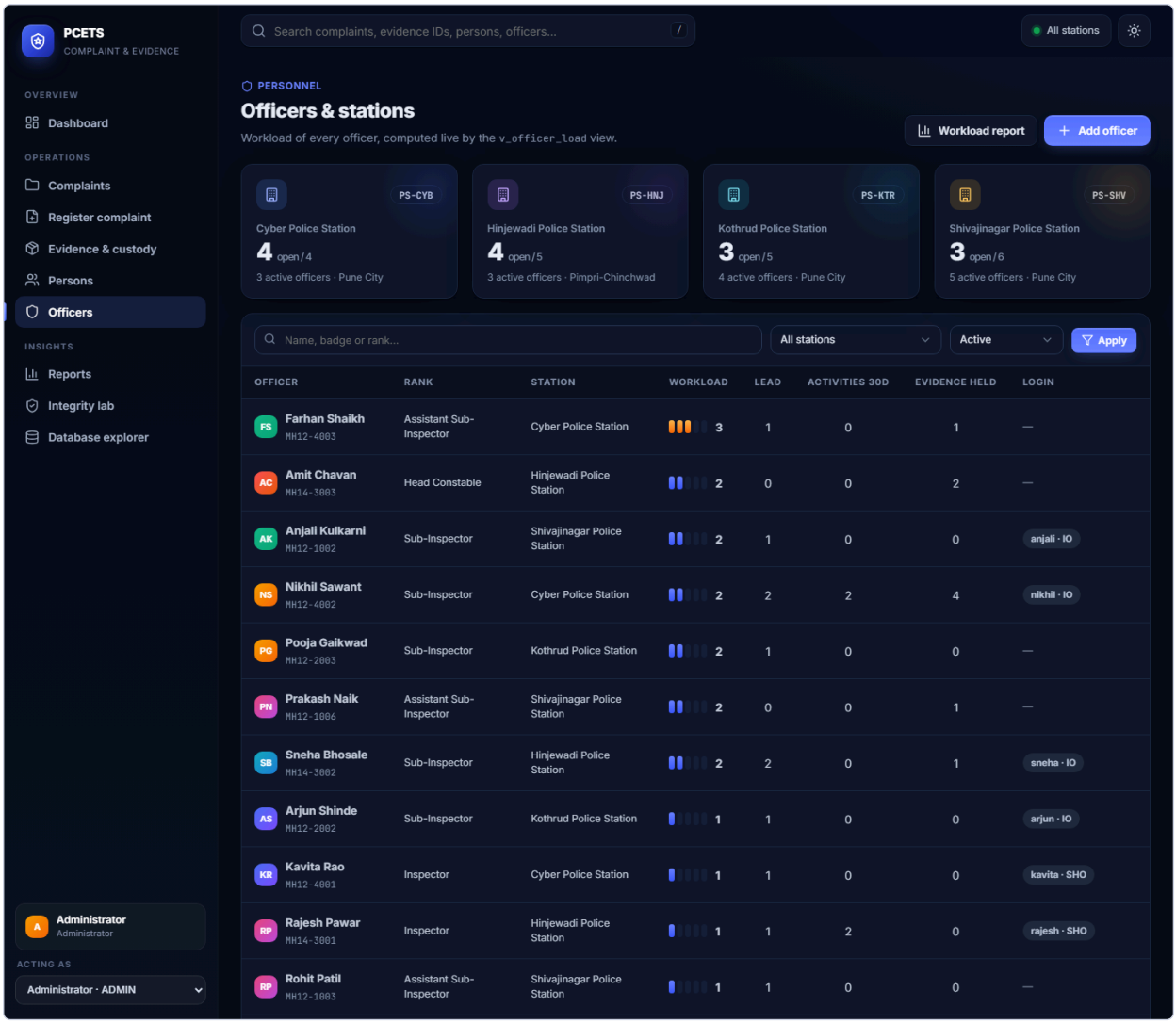


Figure 17. Officers and station summary

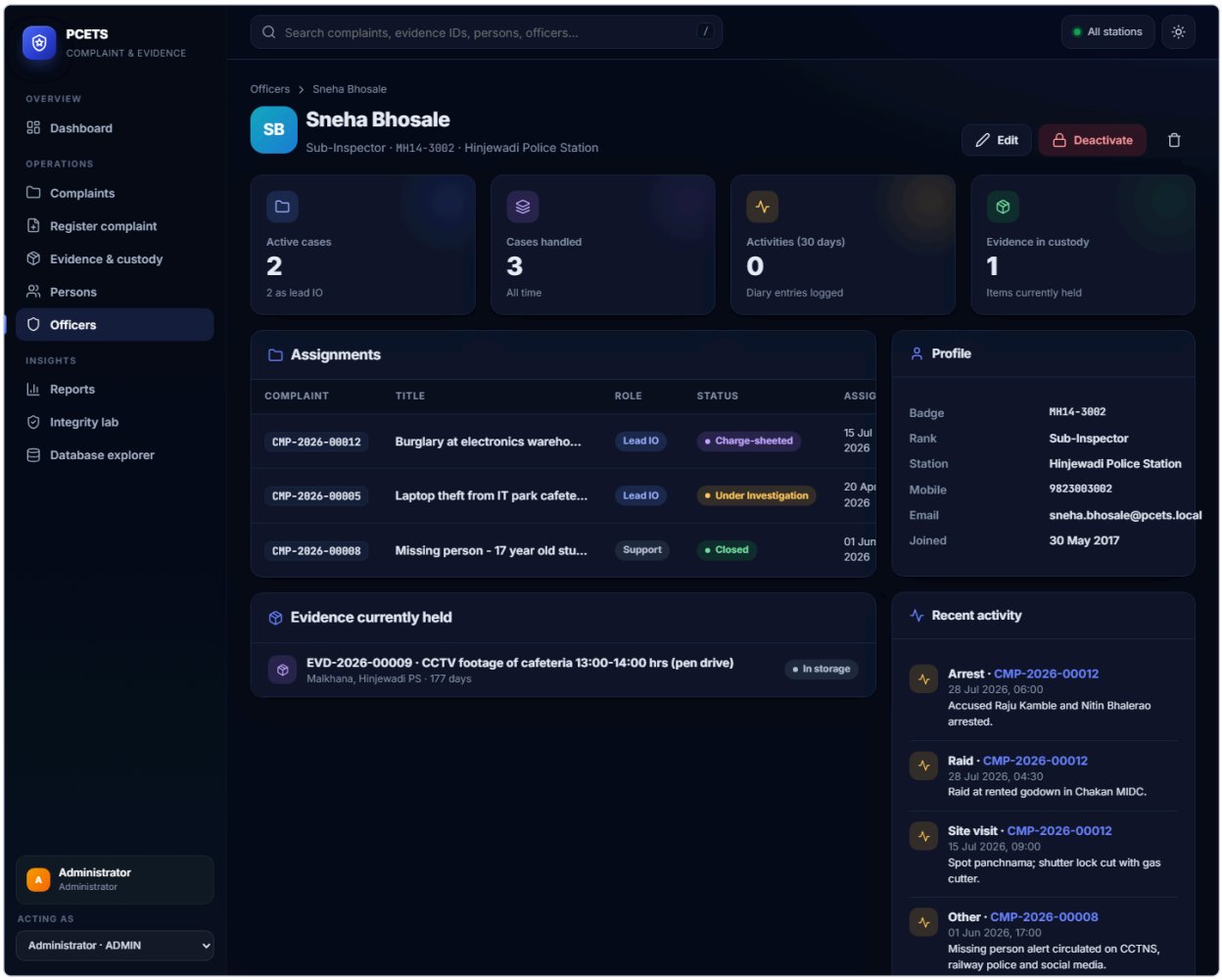


Figure 18. Officer workload profile

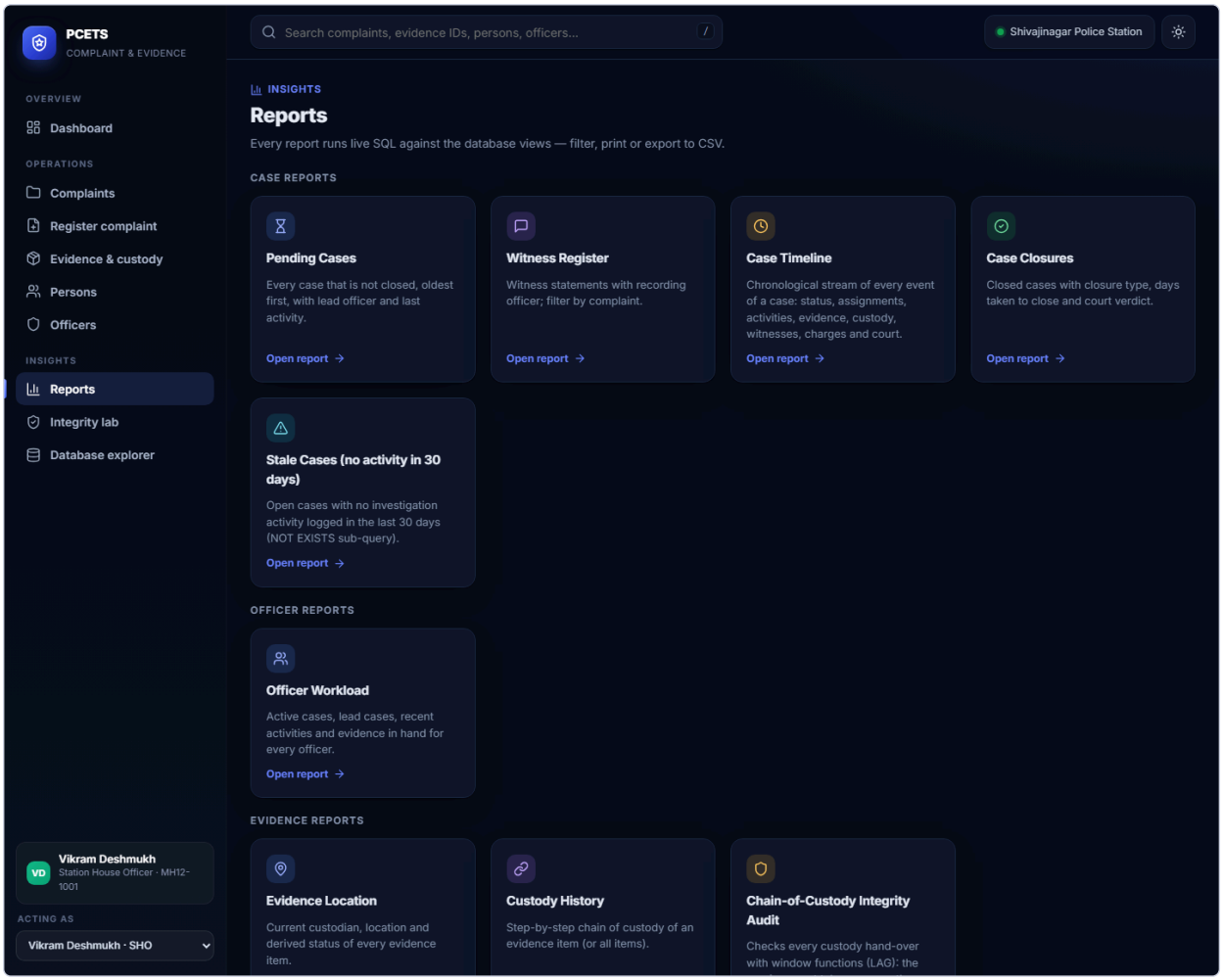


Figure 19. Reports hub

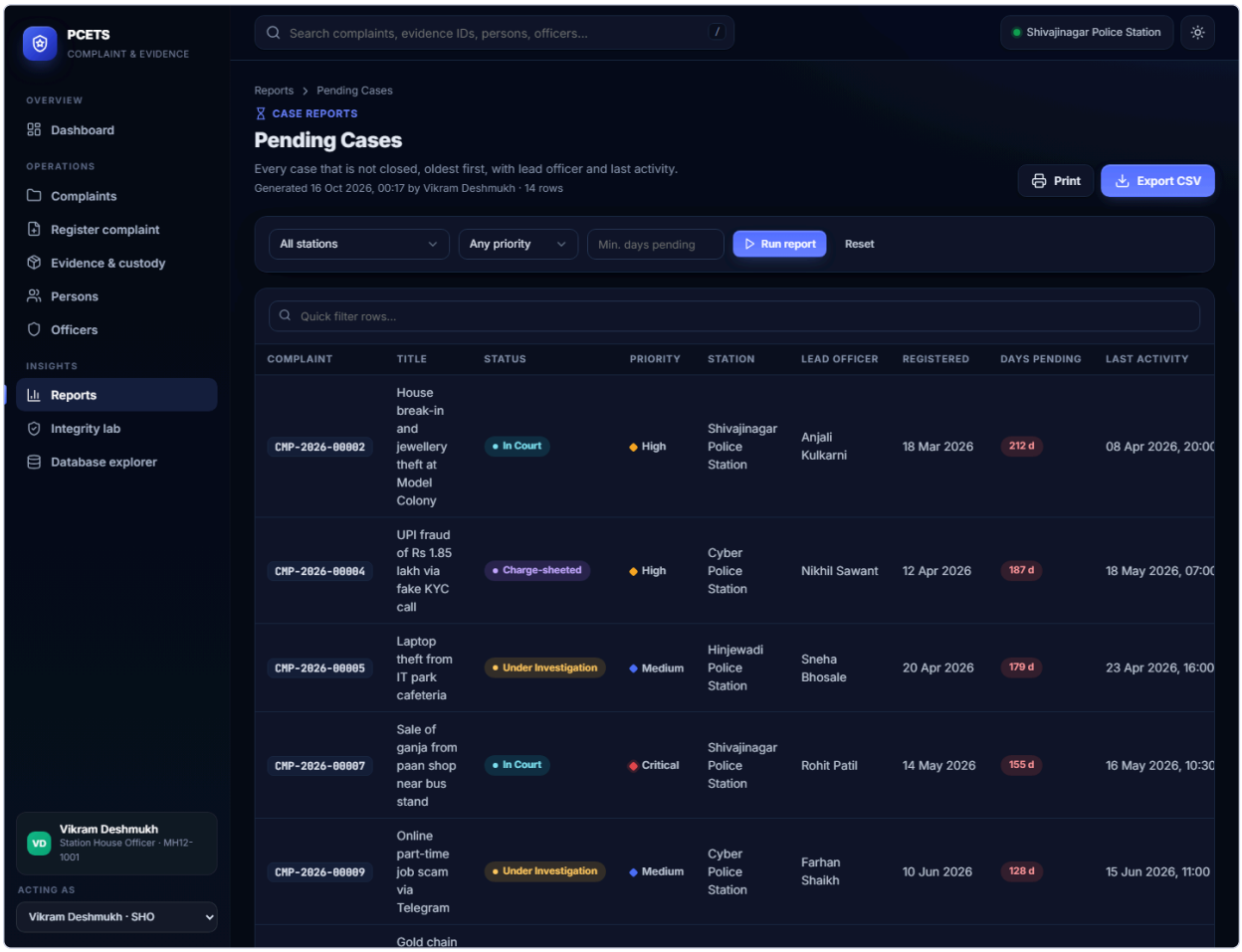


Figure 20. Report — pending cases

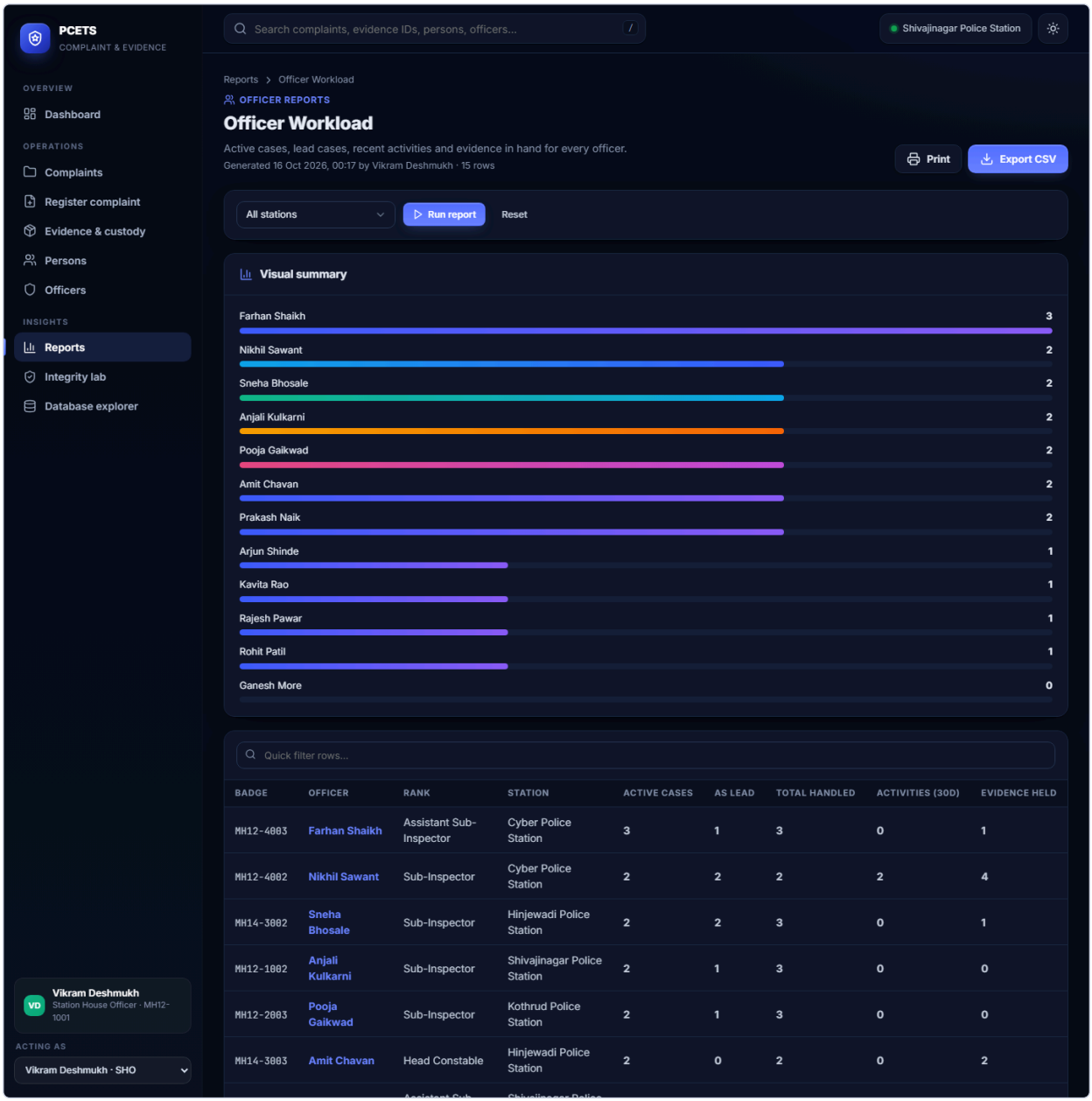


Figure 21. Report — officer load

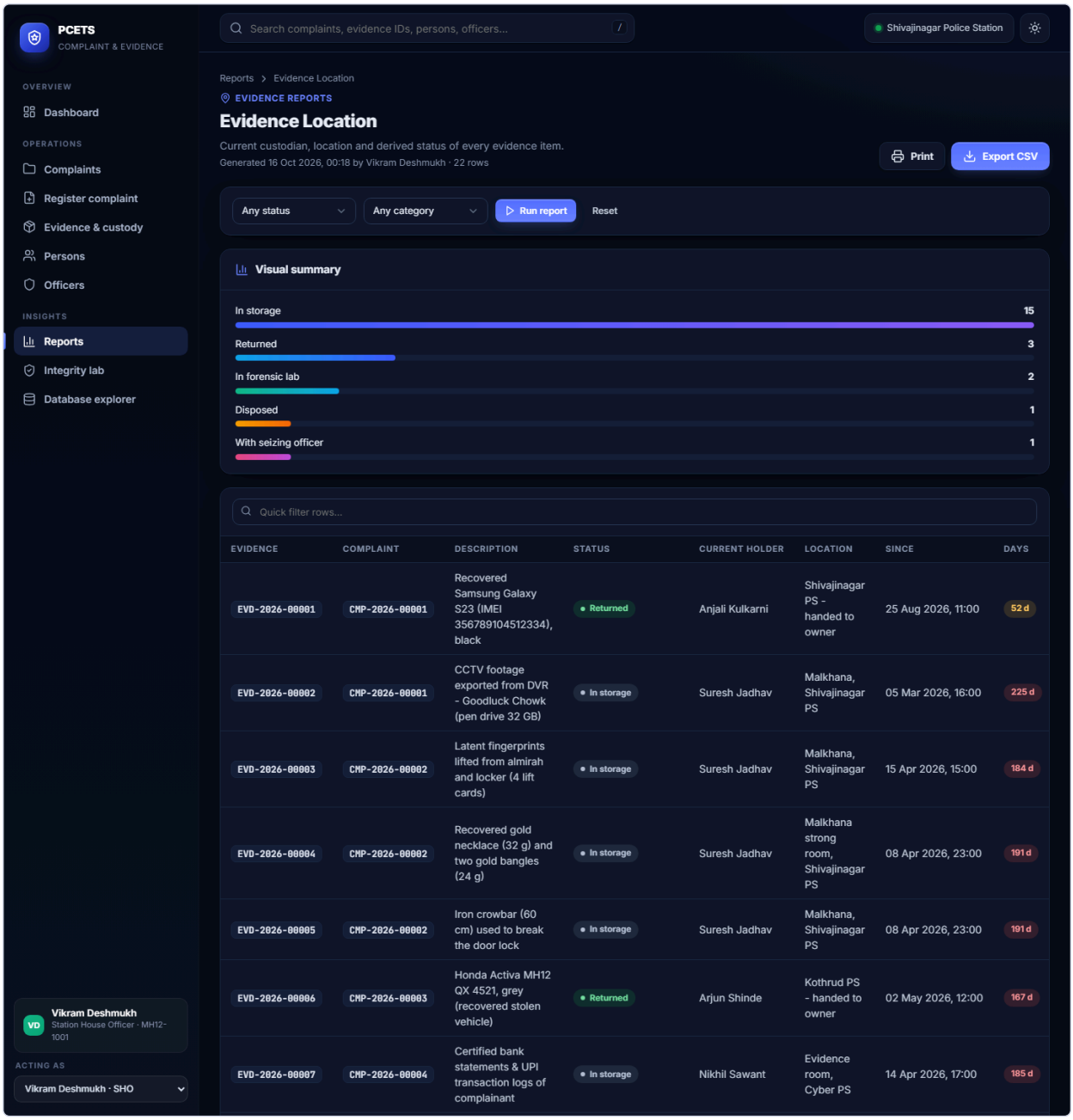


Figure 22. Report — evidence location

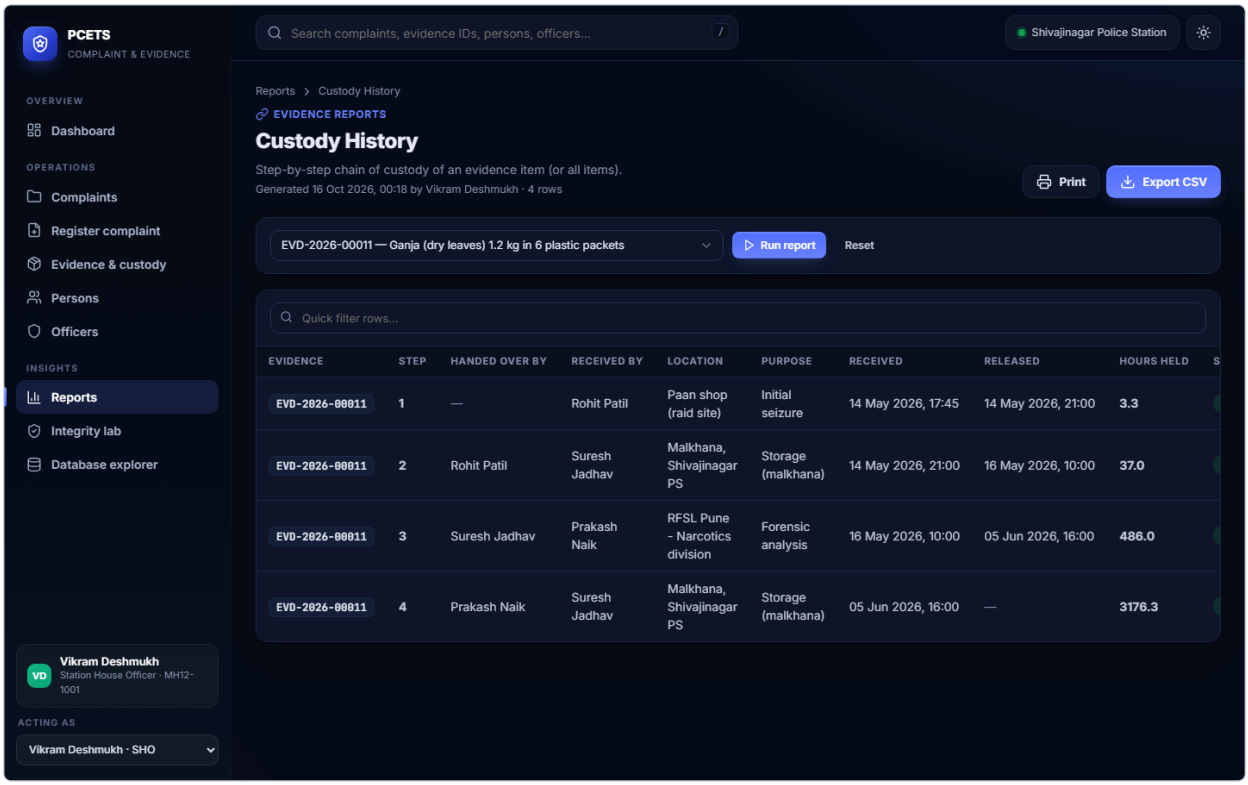


Figure 23. Report — custody history

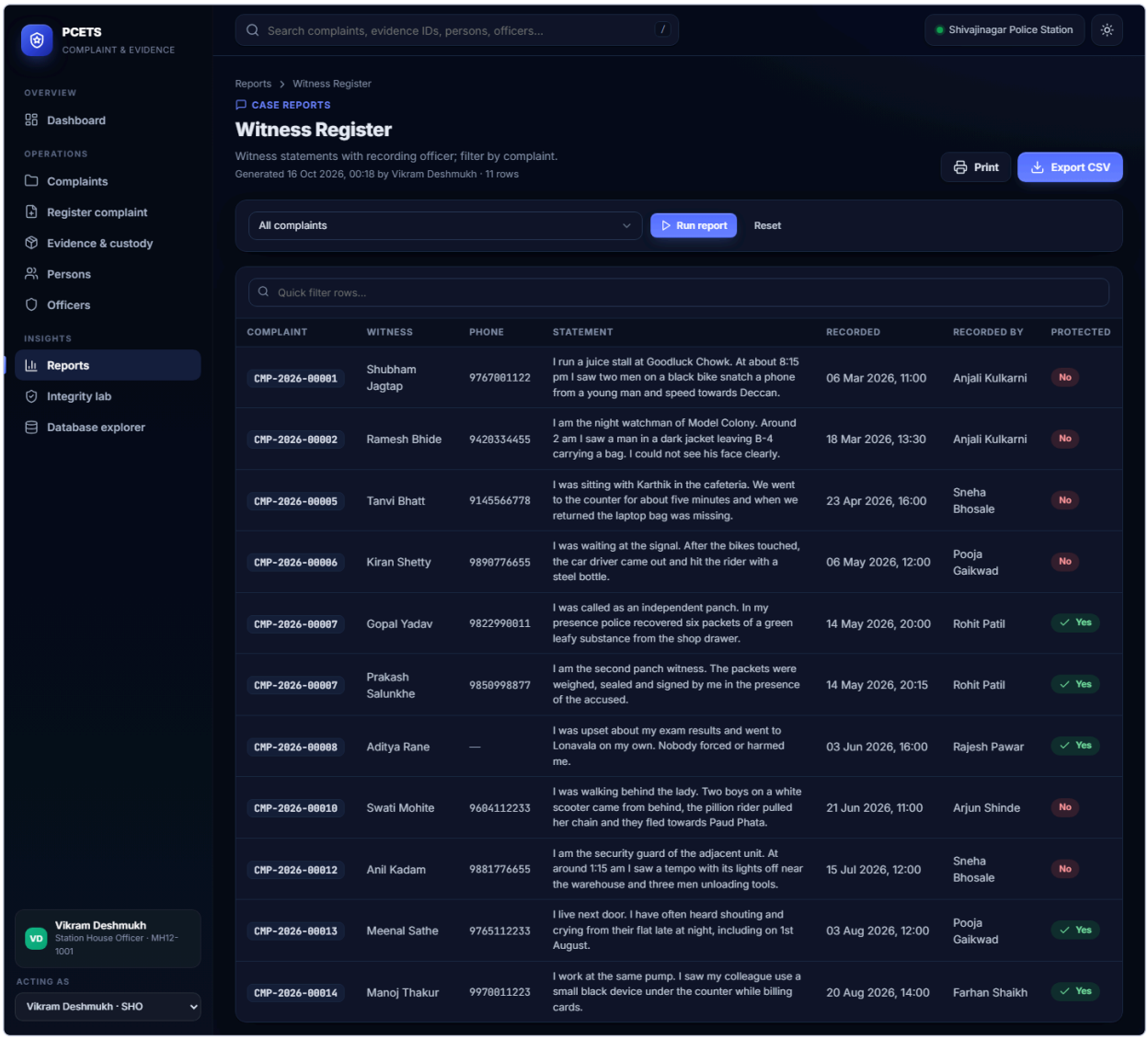


Figure 24. Report — witnesses

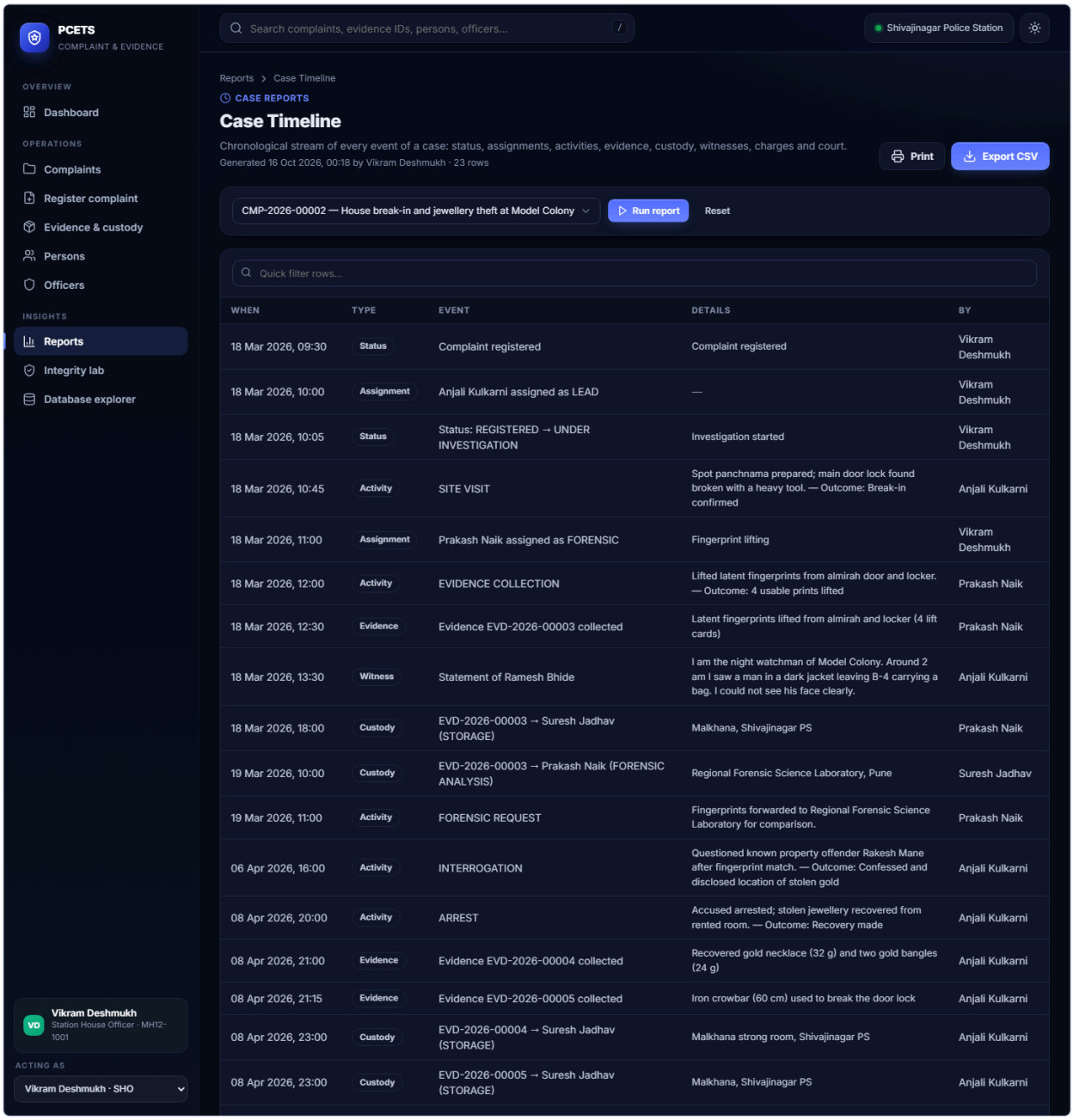


Figure 25. Report — case timeline

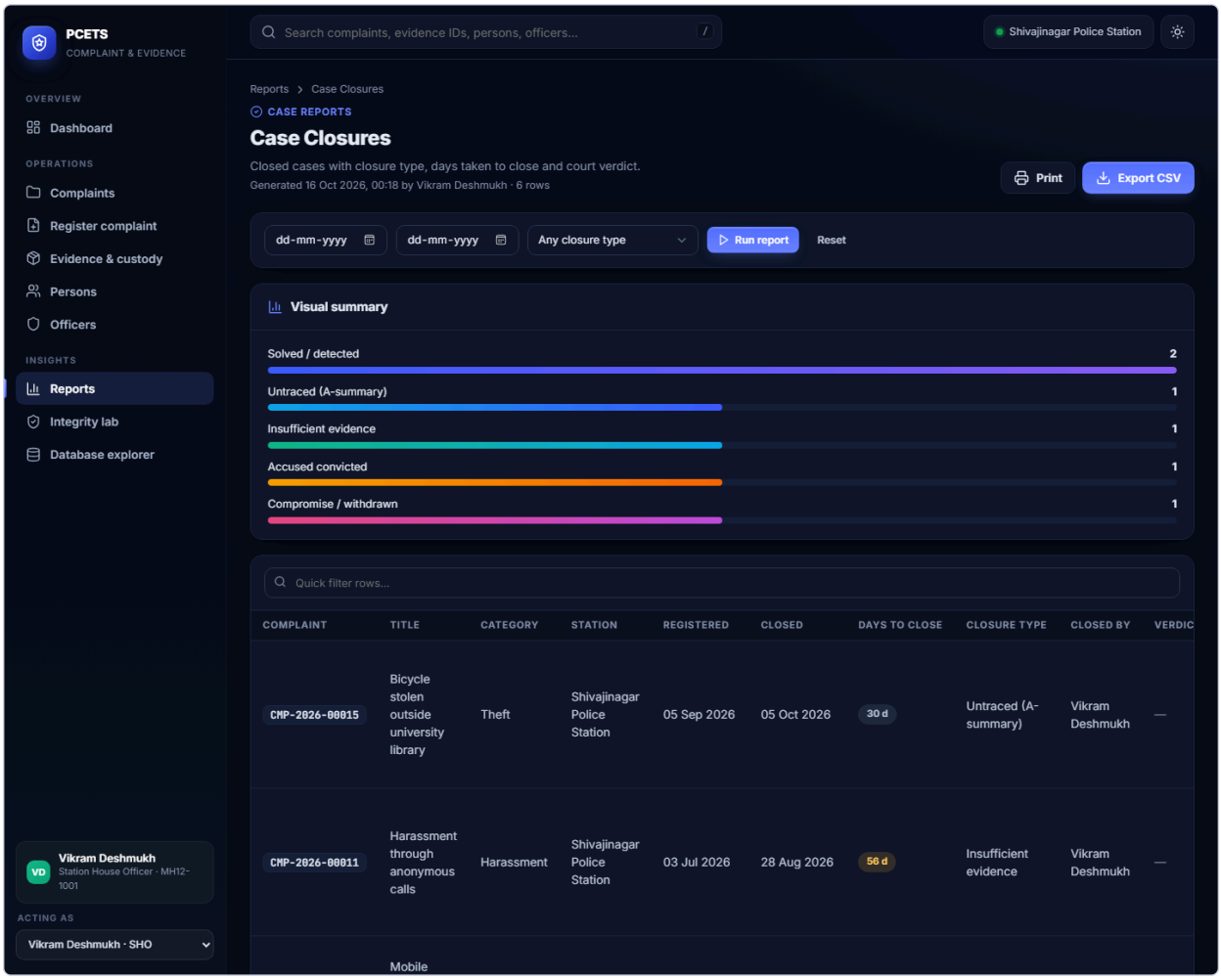


Figure 26. Report — closures

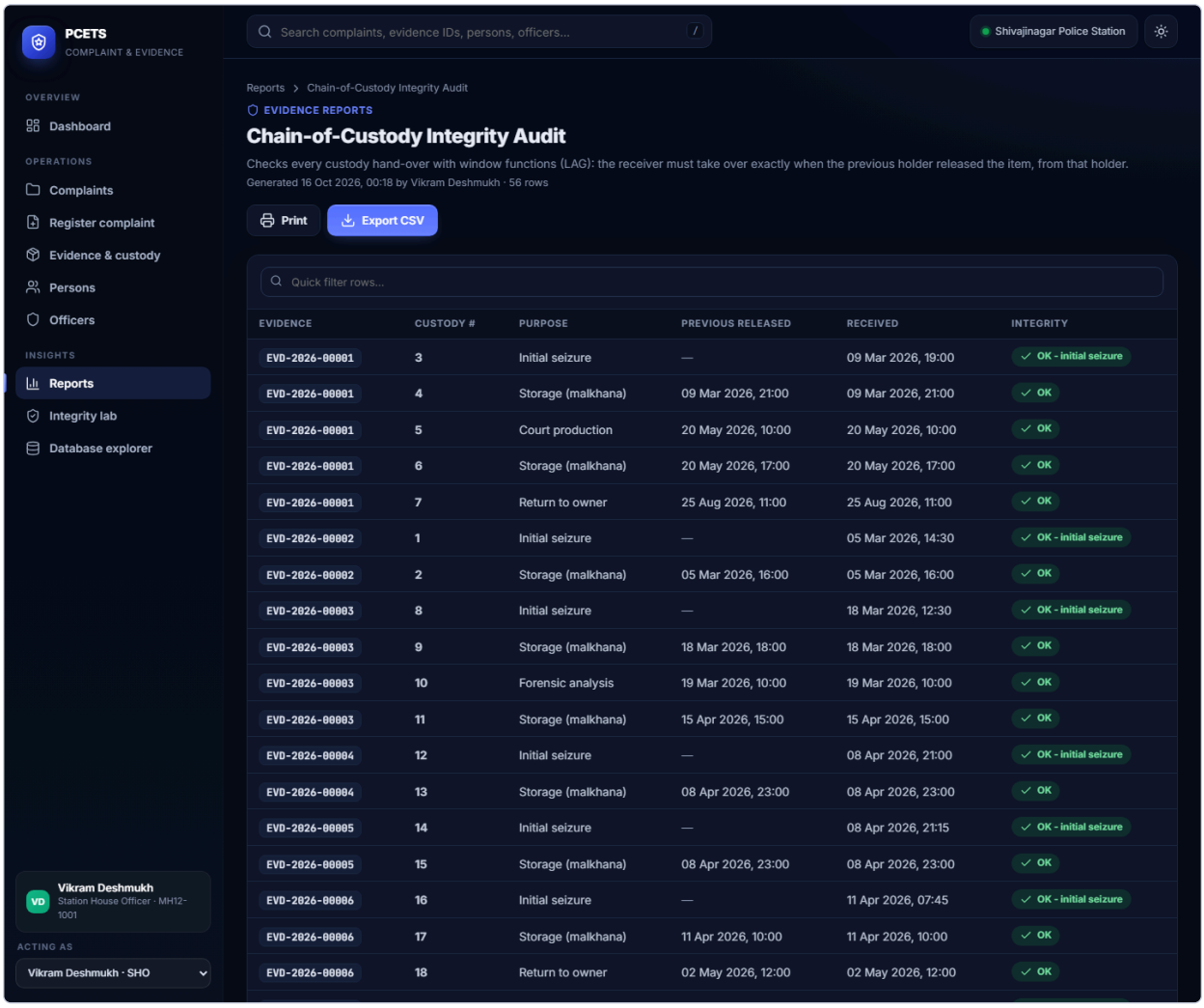


Figure 27. Report — chain-of-custody integrity audit (window functions)

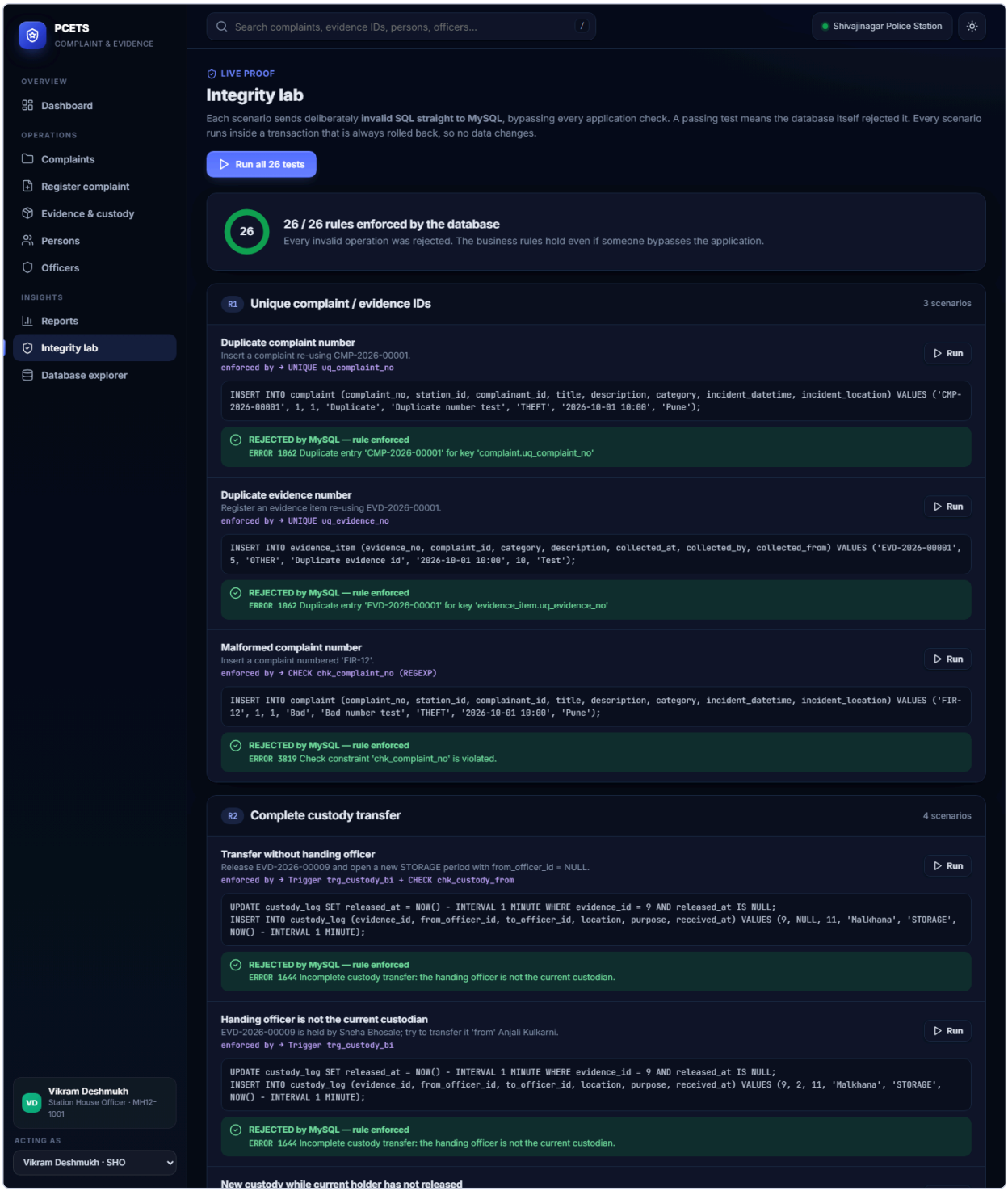


Figure 28. Integrity lab — every rule rejected by MySQL

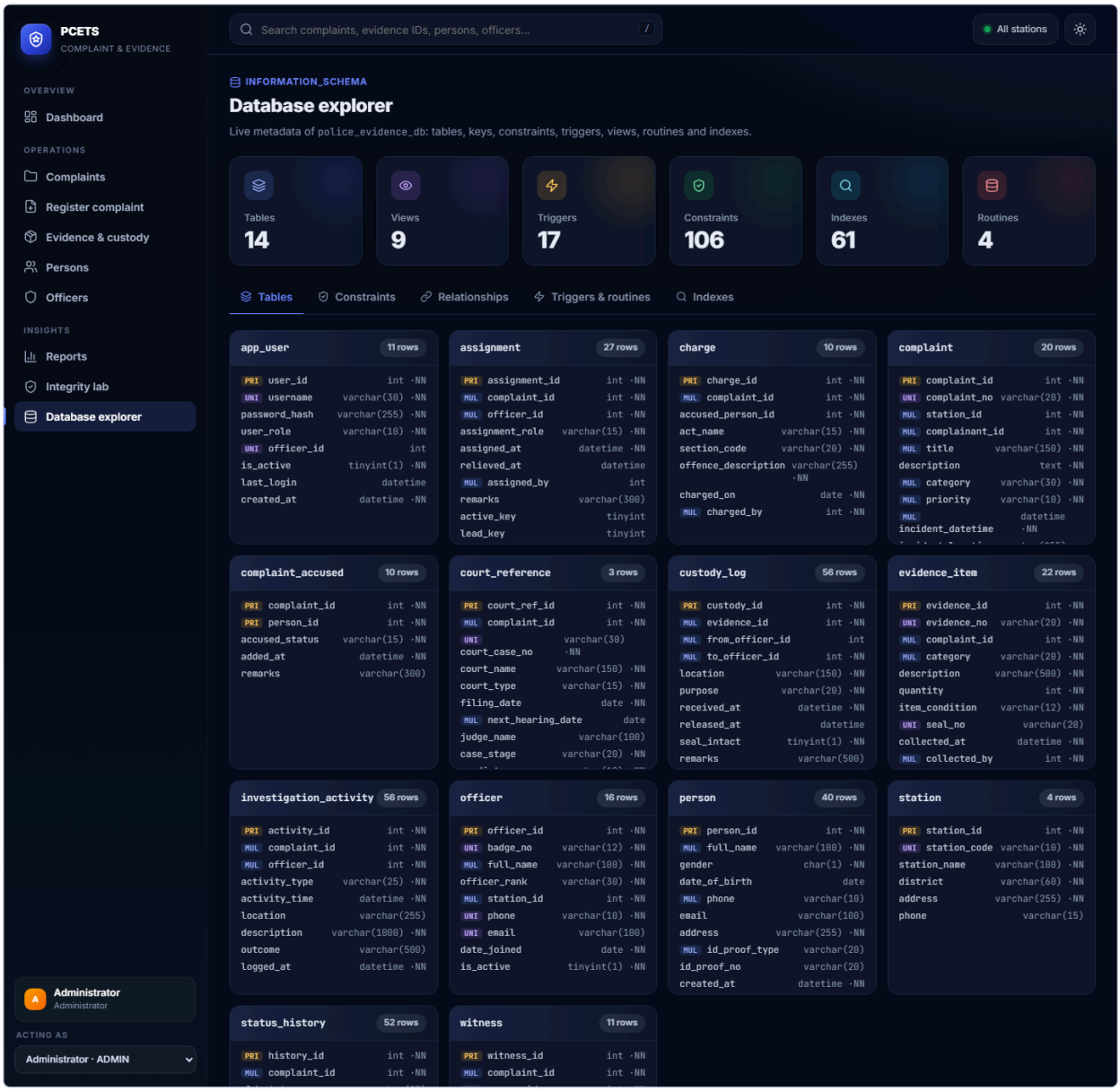


Figure 29. Database explorer (information_schema)

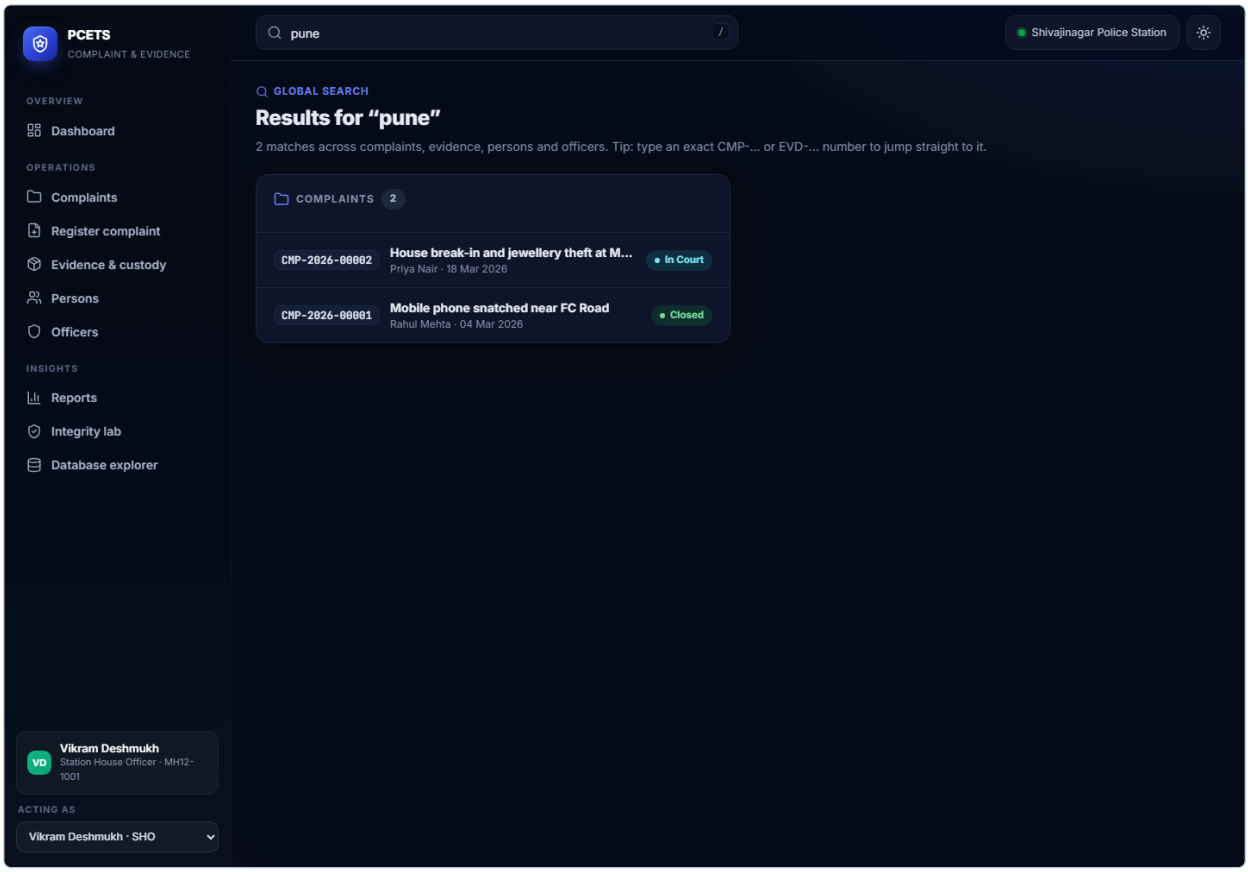


Figure 30. Global search

Print / Save as PDF

CASE FILE — SHIVAJINAGAR POLICE STATION

Pune City · Police Complaint & Evidence Tracking System

CMP-2026-00002
Printed 16 Oct 2026, 00:18

Title	House break-in and jewellery theft at Model Colony
Category / Priority	Burglary / High
Status	**In Court**
Incident	18 Mar 2026, 02:00 at B-4, Model Colony, Shivajinagar, Pune
Registered	18 Mar 2026, 09:30 by Vikram Deshmukh
Complainant	Priya Nair, 9890122334 — B-4, Model Colony, Shivajinagar, Pune 411016

NARRATIVE

Unknown persons broke the main door lock while the family was away and stole gold jewellery worth approx. Rs 4.2 lakh from the bedroom almirah.

INVESTIGATION TEAM

OFFICER	ROLE	ASSIGNED	RELIEVED
Anjali Kulkarni (MH12-1002)	Lead IO	18 Mar 2026, 10:00	—
Prakash Naik (MH12-1006)	Forensic	18 Mar 2026, 11:00	—

ACCUSED & CHARGES

ACCUSED	STATUS	CHARGES
Rakesh Mane	On bail	BNS 331(4) — House-breaking by night BNS 305 — Theft in a dwelling house

EVIDENCE & CURRENT CUSTODY

EVIDENCE	DESCRIPTION	SEAL	STATUS	HELD BY / LOCATION
EVD-2026-00003	Latent fingerprints lifted from almirah and locker (4 lift cards)	SL-SHV-0003	In storage	Suresh Jadhav — Malkhana, Shivajinagar PS
EVD-2026-00004	Recovered gold necklace (32 g) and two gold bangles (24 g)	SL-SHV-0004	In storage	Suresh Jadhav — Malkhana strong room, Shivajinagar PS
EVD-2026-00005	Iron crowbar (60 cm) used to break the door lock	SL-SHV-0005	In storage	Suresh Jadhav — Malkhana, Shivajinagar PS

WITNESSES

WITNESS	RECORDED	STATEMENT
Ramesh Bhide	18 Mar 2026, 13:30	I am the night watchman of Model Colony. Around 2 am I saw a man in a dark jacket leaving B-4 carrying a bag. I could not see his face clearly.

COURT

CASE NO.	COURT	FILED	NEXT HEARING	STAGE / VERDICT
RCC-0782-2026	Judicial Magistrate First Class, Shivajinagar	10 May 2026	28 Oct 2026	Hearing / Pending

CHRONOLOGY

WHEN	EVENT	BY
18 Mar 2026, 09:30	**Complaint registered** Complaint registered	Vikram Deshmukh
18 Mar 2026, 10:00	**Anjali Kulkarni assigned as LEAD**	Vikram Deshmukh
18 Mar 2026, 10:05	**Status: REGISTERED → UNDER INVESTIGATION** Investigation started	Vikram Deshmukh
18 Mar 2026, 10:45	**SITE VISIT** Spot panchnama prepared; main door lock found broken with a heavy tool. — Outcome: Break-in confirmed	Anjali Kulkarni
18 Mar 2026, 11:00	**Prakash Naik assigned as FORENSIC** Fingerprint lifting	Vikram Deshmukh
18 Mar 2026, 12:00	**EVIDENCE COLLECTION** Lifted latent fingerprints from almirah door and locker. — Outcome: 4 usable prints lifted	Prakash Naik
18 Mar 2026, 12:30	**Evidence EVD-2026-00003 collected** Latent fingerprints lifted from almirah and locker (4 lift cards)	Prakash Naik
18 Mar 2026, 13:30	**Statement of Ramesh Bhide**	Anjali Kulkarni

Figure 31. Printable case file

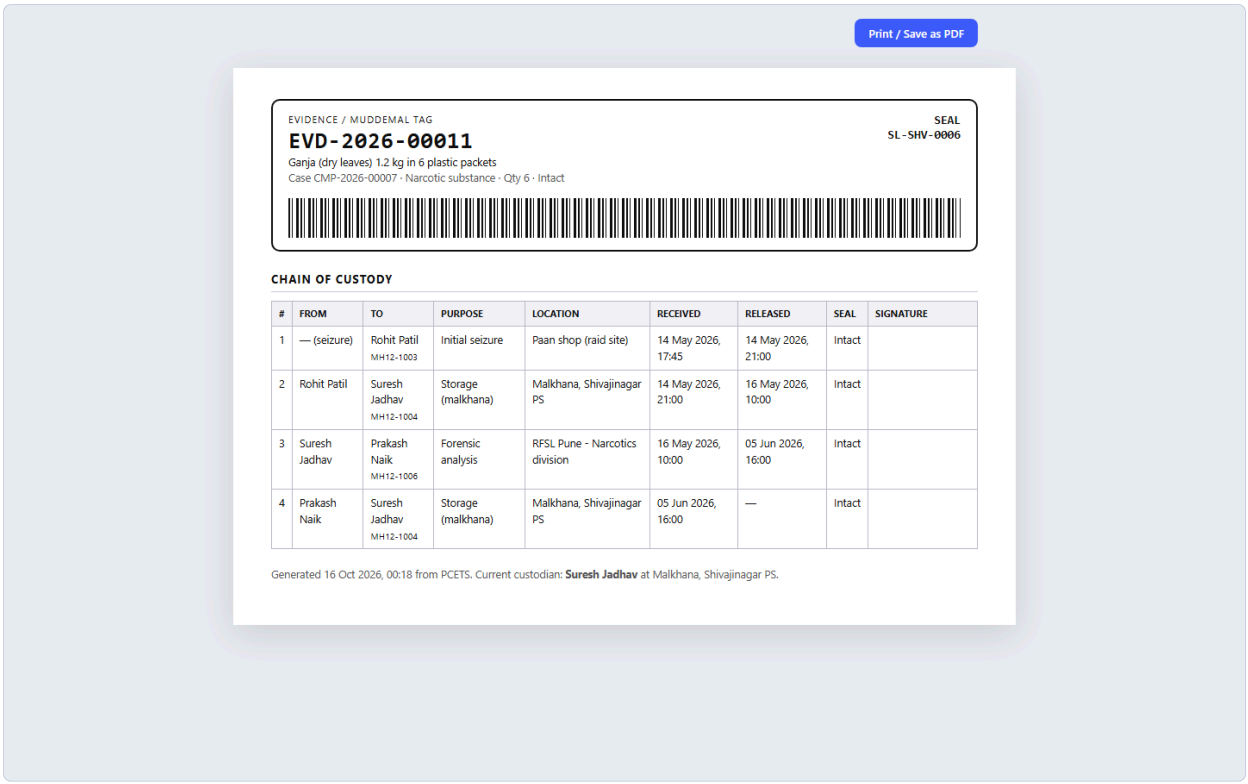


Figure 32. Printable custody sheet / evidence tag

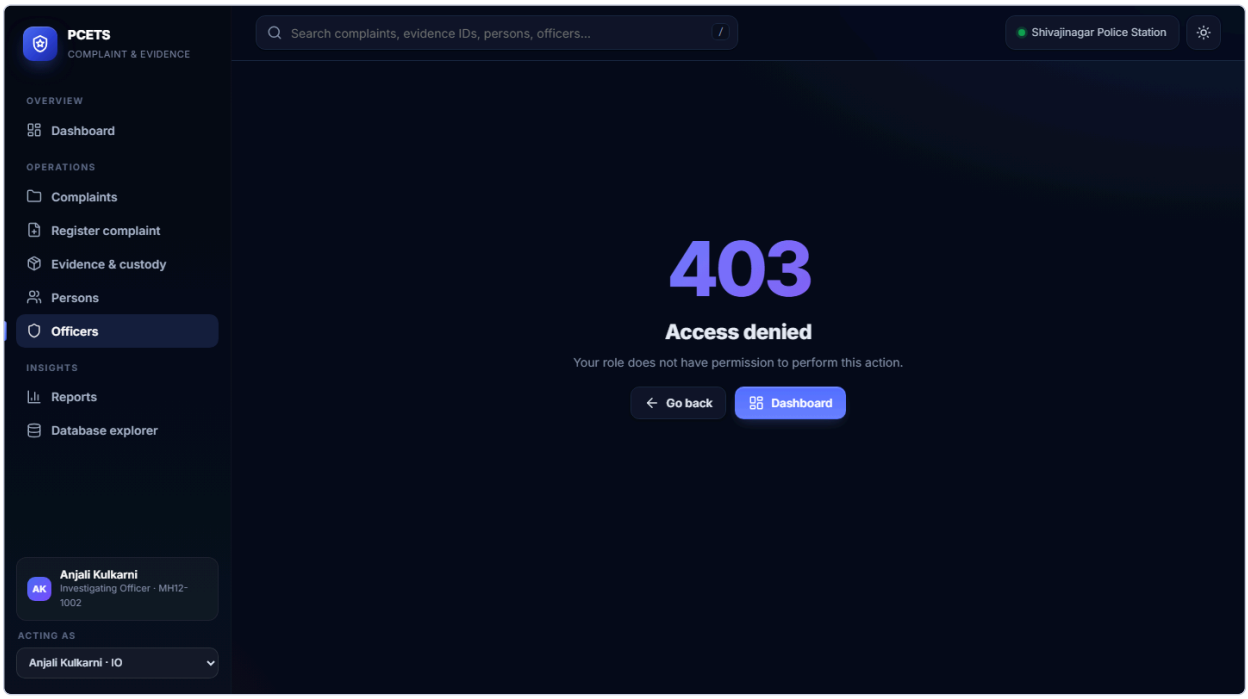


Figure 33. Role-based access control — acting as an IO, adding an officer is denied

16. Conclusion and future scope

PCETS replaces fragmented station registers with a single normalized database whose rules cannot be bypassed: complaint and evidence identifiers are unique, every custody hand-over is complete and gap-free, custody periods never overlap, case status follows a valid workflow with a permanent audit trail, and a case cannot be closed without its closure details. The Flask application exposes all required operations — complaint registration, officer assignment, activity logging, evidence registration, custody transfer, witness entry, status update and reporting — with validation, transactions, role-based security and a modern responsive interface. 43 automated tests and 26 database-level scenarios verify the behaviour.

Future scope

- Attachments (photos, PDFs of statements) stored with hash values to prove evidence files are unaltered.
- QR codes on evidence tags scanned at hand-over; SMS/e-mail alerts for hearing dates and long lab stays.
- Integration with CCTNS and e-Courts APIs; multi-district partitioning; read replicas for analytics.

Appendix A. Installation and running

```
1. Install MySQL 8.0 and Python 3.10+
2. pip install -r requirements.txt
3. Edit config.ini  (MySQL host / user / password)
4. python setup_db.py          # creates police_evidence_db, loads schema, triggers, views and sample data
5. python run.py               # open http://127.0.0.1:5000
6. python -m unittest -v tests.test_app     # automated tests (uses a separate test database)

No login: the app opens as admin; switch to vikram (SHO), anjali (IO), suresh (Custodian)… with "Acting as" in the sidebar
SQL only:     run sql/00 … sql/04 in MySQL Workbench; sql/05 = report queries; sql/06 = negative tests (mysql --force)
```

Appendix B. Project structure

```
Police Evidence Tracking/
├── config.ini · requirements.txt · run.py · setup_db.py · README.md
├── sql/       00_create_database · 01_schema · 02_triggers_procedures · 03_views · 04_seed_data · 05_report_queries · 06_constraint_tests
├── police_app/
│   ├── app.py (factory, filters, error pages) · auth.py (acting user, RBAC, CSRF) · db.py (pool, transactions, error translation)
│   ├── validators.py · constants.py
│   ├── services/  complaint · evidence · person · officer · report · dashboard · integrity
│   ├── routes/    main · complaints · evidence · persons · officers · reports · admin
│   ├── templates/ (Jinja2)  ·  static/css/app.css  ·  static/js/app.js
├── tests/test_app.py
├── tools/  capture_screens.py · build_report.py · report_template.html
└── docs/   Project_Report.html/.pdf · TEST_CASES.md · er_diagram.mmd/.png · screenshots/
```

Appendix C. Sample data

Table	Rows	Table	Rows
app_user	11	assignment	27
charge	10	complaint	20
complaint_accused	10	court_reference	3
custody_log	56	evidence_item	22
investigation_activity	56	officer	16
person	40	station	4
status_history	52	witness	11

The sample data describes 20 realistic 2026 cases across 4 Pune stations (robbery, burglary, cyber fraud, narcotics, missing person, domestic violence …) at every stage of the workflow, 22 evidence items with 56 custody periods (including forensic lab visits, court production, return to owner and disposal) — all inserted through the triggers, which proves the rules are mutually consistent.